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FULL LENGTH ARTICLE

Aerodynamic analysis of rotor-to-rotor interactions in different octocopter configurations



Aqib AZIZ, Yongjie SHI *, Yang LIU, Guohua XU

National Key Laboratory of Rotorcraft Aeromechanics, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China

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KEYWORDS

Rotor-to-rotor interaction;
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Octocopter configurations;
Aerodynamically efficient;
Hover;
Forward flight

Abstract Rotor-to-rotor interaction among neighboring rotors of a multirotor has great significance for aerodynamically efficient multirotor design. Current research is conducted to analyze aerodynamic performance of different octocopter configurations amid hover and forward flight. Conventional and coaxial configurations are studied and a hybrid configuration is also proposed to rectify the disadvantages associated with the earlier two. Comparison is carried out for the aforementioned configurations along with comparison of coaxial and hybrid octocopters with bigger diameter rotors in the same confined space for high thrust requirement missions. Vertical spacing of coaxial configuration is also studied. Virtual Blade Method (VBM) is considered herein due to its great computational efficiency. The results show that there are 11.89% and 14.22% loss in thrust for coaxial octocopter compared to conventional and hybrid configurations with normal size rotors and 15.61% loss compared to hybrid configuration with bigger rotors in hover, whereas coaxial square configuration performs the worst in forward flight with a lift loss of 9.1%, 14.77% and 18.8% compared to coaxial diamond, conventional and hybrid configurations with normal size rotors and 9.96% and 17.82% loss compared to coaxial diamond and hybrid configurations with bigger rotors. Combined FM shows that hybrid configuration outperforms other octocopter configurations in overall aerodynamic performance.
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1. Introduction

Development of small Unmanned Aerial Vehicles (UAVs) is being carried out at an amazingly rapid pace in recent years for a wide range of applications.[1] Their common civilian applications include field site surveillance, package delivery, making videos, personal entertainment, and disease control. New and novel designs of such aircrafts are presented. A great shift in interest towards eVTOL technology is observed in recent years. Thipphavong et al.[2] provided a detailed description of emergent and early expanded operations of NASA regarding new aircraft designs for the developments in Urban Air Mobility (UAM). Work of Zhang et al.[3] provides a review of the UAM management from perspectives of flow control, traffic services, airspace planning and infrastructure along with the challenges faced by aerospace industry and future develop-

* Corresponding author.
E-mail address: shiyongjie@nuaa.edu.cn (Y. SHI).
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ments in this regard. Warwick et al.[4] and Rajendran et al.[5] highlight the importance of eVTOL technology for developing air taxi service for UAM and also provided a critical review of these developments. Manufacturers including Uber,[6] Rolls Royce,[7] Airbus and Volocopter GmbH have unveiled various prototypes or conceptual designs for UAVs. Joshi[8] worked on flapping type wing miniature UAVs with different shapes of the wings and various flow conditions utilizing CFD tools for the analysis purpose and highlighted one additional merit of UAVs, being small in size which allows them to move in confined spaces. Houghton and Hoburg[9] conducted early research on monocopter UAV operations in confined spaces and indoor operations employing CFD analysis, parametric modelling and flight tests. Their study mentions that similar to traditional helicopters, these small UAVs also generate thrust by a constant rotational motion of the rotor blades with great capability of indoor and outdoor operations in confined spaces. The unique ability of vertical takeoff and landing has attracted many engineers and researchers. Otsuka and Nagatani[10] presented an overlapping configuration of rotors to reduce the thrust loss due to rotor-wake interactions in hover and also highlighted the fact that these small UAVs are able to perform several tasks by just controlling the rotational speeds of their rotors. Electric Vertical Takeoff and Landing (eVTOL) vehicles are able to operate at much higher speeds than the corresponding traditional helicopters with a main rotor and a tail rotor having limitations of blade tip velocity. This advantage of eVTOL vehicles is also seen in research of Nugroho et al.[11] and Chahba et al.,[12] who performed analysis at different high RPMs for rotors of such multirotor which is not feasible for helicopter rotors.

At present, in addition to some customized configurations with different rotor arrangements, tricopters, quadcopters, hexacopters and octocopters are commonly available in market.[13–16] Different configurations have different thrusting capability, operation time and range depending on the type of given task.[17–19] Although multirotor UAVs have their promising applications and significant benefits, there are a few challenges associated as well which require technical improvements to enhance their aerodynamic performance. Ko and Lee[20] worked on the rotor-wake interactions of multirotor in forward flight and Lakshminarayan and Baeder[21] performed analysis of rotor wake interaction of coaxial rotors for multirotor in hover and both researches conclude that unlike traditional helicopters, the rotors of small multirotor UAVs are significantly affected by the trailed wake system both in hover and forward flight since each rotor remains in the proximity of such wake, especially in hovering state. Currently available small multirotor UAVs have operating time of less than an hour, which indicates a need of improvements to enhance this factor. One effective way to achieve this goal is to enhance aerodynamic efficiency of rotors by reducing interaction with neighboring rotors. Aeroacoustics noise is another phenomena besides aerodynamic performance which is also linked comprehensively with rotor-to-rotor interaction in small multirotor UAVs. Different multirotor configurations have different extent of rotor-wake interaction and developing new configurations which can minimize such interactions is a must. Therefore, an accurate prediction of rotor-to-rotor interaction is quite essential for aerodynamically efficient design of small multirotor UAVs.

Even though various studies are being conducted on small multirotor UAVs, very limited literature is available for rotor-to-rotor interaction and its influence on aerodynamic efficiency of rotor in different small multirotor UAV configurations especially in forward flight. It might be because of the fact that rotor-to-rotor interaction is more evident in hover than in forward flight case. Except for Yoon et al.[22] who performed computational analysis on aerodynamic performance of multirotor in a quadcopter configuration and concluded that vertical forces of the quadcopter are significantly influenced by rotor interaction in hovering state. Many scholars[23–26] conducted analysis on aerodynamic performance of coaxial rotor configuration in different flight conditions employing CFD methods, since coaxial rotors in hovering state have the most severe wake interaction. Russell et al.[27] carried out a comprehensive study through quantifying thrusts of five multirotor UAVs at different vehicle attitudes, rotor rotational speeds and wind speeds. Research is also focused on developing new tools for accurate and fast results prediction for multirotor studies involving very heavy grids. Tugnoli et al.[28] developed a tool called DUST which is an open source code developed with MIT license for the aforementioned purpose, Caprace et al.[29] studied the quadcopter in forward flight with focus on the influence on aerodynamic performance of multirotor due to interaction of airframe of the vehicle with the wake of rotors using vortex particle mesh method which is another CFD based method used widely for wake studies and Zhang et al. [30] studied high fidelity aerodynamic and aeroacoustics modeling processes which support preliminary design of large eVTOL vehicles, employing CFD tools. They also focused on aerodynamic interference effects between rotors and other components. Yang et al.[31] performed simulations and experimental analysis on an isolated multirotor propeller in forward flight at different advance ratios to understand thrust and torque characteristics.

In present research, rotor-to-rotor interaction in octocopters is studied employing CFD tools to understand the influence of such interaction on aerodynamic performance of the octocopters in conventional and coaxial configurations. Thrust, torque as well as figure of merit for each configuration is provided to understand the aerodynamic behavior of each configuration. The mathematical representation of the combined figure of merit for eight rotors of the different octocopter configurations is given in the study of Shukla and Komerath[32] and mentioned in Eq. (1).

$$\mathrm{FM}_{\mathrm{combined}} = \frac{(C_{T1}^{\frac{3}{2}} + C_{T2}^{\frac{3}{2}} + C_{T3}^{\frac{3}{2}} + C_{T4}^{\frac{3}{2}} + C_{T5}^{\frac{3}{2}} + C_{T6}^{\frac{3}{2}} + C_{T7}^{\frac{3}{2}} + C_{T8}^{\frac{3}{2}})/\sqrt{2}}{C_{Q1} + C_{Q2} + C_{Q3} + C_{Q4} + C_{Q5} + C_{Q6} + C_{Q7} + C_{Q8}} \tag{1}$$

where C_{Ti} shows thrust coefficient for each rotor and C_{Qi} shows torque coefficient for each rotor and 'i' shows the numbering for each rotor. Since power is a product of torque and Revolutions Per Minute (RPM), and RPM is kept the same for each rotor, therefore both torque coefficient and power coefficient can be used in the denominator of FM equation.

Rotor-to-rotor interaction in coaxial octocopter due to different vertical spacing is also analyzed. Moreover, a hybrid octocopter configuration is proposed to improve the aerodynamic efficiency of octocopters by minimizing rotor-to-rotor interaction. The proposed hybrid configuration is a hybrid of conventional and coaxial configurations. The arrangements

of all rotors in this configuration is same as conventional octocopter but four alternating rotors are moved downwards just like two layers in coaxial configuration. The difference is that the lower four rotors of hybrid configuration are not placed exactly below the upper layer of rotors as in coaxial and they have no overlap. A comprehensive study is conducted to compare abovementioned configurations in hover and forward flight cases. In addition to this, research is also performed to understand thrust and torque performance of coaxial and hybrid configurations with larger diameter rotors for high thrust requirement missions, since coaxial and hybrid configurations provide the opportunity to expand the rotor diameter within confined space, which is not feasible in conventional octocopter configuration. Current research will enable the readers to understand which configuration comparatively outperforms the other configurations under hovering and forward flight conditions and the magnitude of influence of rotor-to-rotor interaction for each configuration.

2. Numerical methods

The solver employed herein is a three-dimensional unsteady Computational Fluid Dynamics (CFD) solver based on Unsteady Reynolds-Averaged Navier Stokes (URANS) equations as the governing equations with k-ω-SST (Shear Stress Transport) as the turbulence model as shown in Eq. (2).

$$\int_{\Omega} \frac{\partial \boldsymbol{W}}{\partial t} \mathrm{d}\Omega + \oint_{\partial\Omega} (\boldsymbol{F}_c - \boldsymbol{F}_v) \cdot \boldsymbol{n} \mathrm{d}s = \int_{\Omega} S \cdot \boldsymbol{n} \mathrm{d}\Omega \tag{2}$$

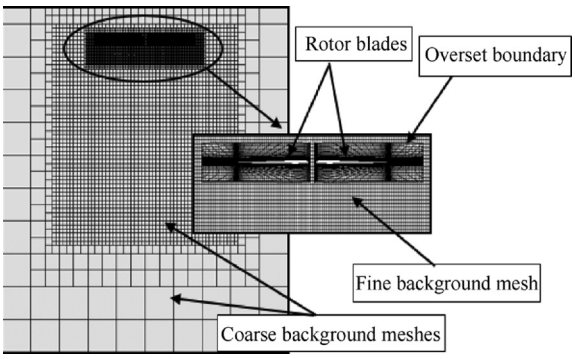


Fig. 1 Overset grid method.

where ds is the cell surface area, $\boldsymbol{W}$ is the vector showing conserved variables and $\boldsymbol{F}_c$ and $\boldsymbol{F}_v$ are the inviscid and viscous flux respectively. Star-CCM+ is utilized as fluid solver, which is considered as a general finite volume, and unstructured multi-physics solver. All computations are carried out on 2 × AMD EPYC 7742 64 Cores 2.4 GHz CPUs with a coupled flow solver. The solver consists of an unsteady double-time-step method having temporal discretization accuracy of second order. Second order upwind scheme is employed for spatial discretization and Roe flux difference splitting scheme is utilized to obtain inviscid flux. Parallel computing is utilized involving multicore processors. The methods utilized in present study are overset grid method and virtual blade method.

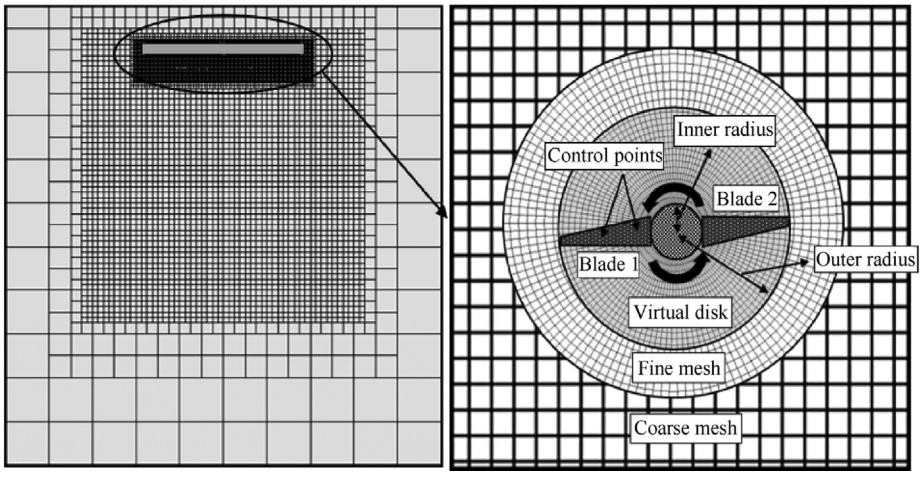


Fig. 2 VBM mesh.

Table 1 Comparison of body fitted grid and VBM grid.

Mesh type	Disc density	Downwash density	Cells for rotor blades	Total cell number (10^6)	Thrust (N)	Difference in thrust (%)	Time for one revolution (h)
Body fitted grid		0.04R	2.86 × 10^6	8.71	2.75		28.26
Coarse mesh	0.025R	0.07R	No cells	1.57	2.85	2.51	1.28
Medium mesh	0.015R	0.04R	No cells	6.41	2.78	1.09	4.95
Fine mesh	0.01R	0.03R	No cells	8.12	2.76	0.36	6.14

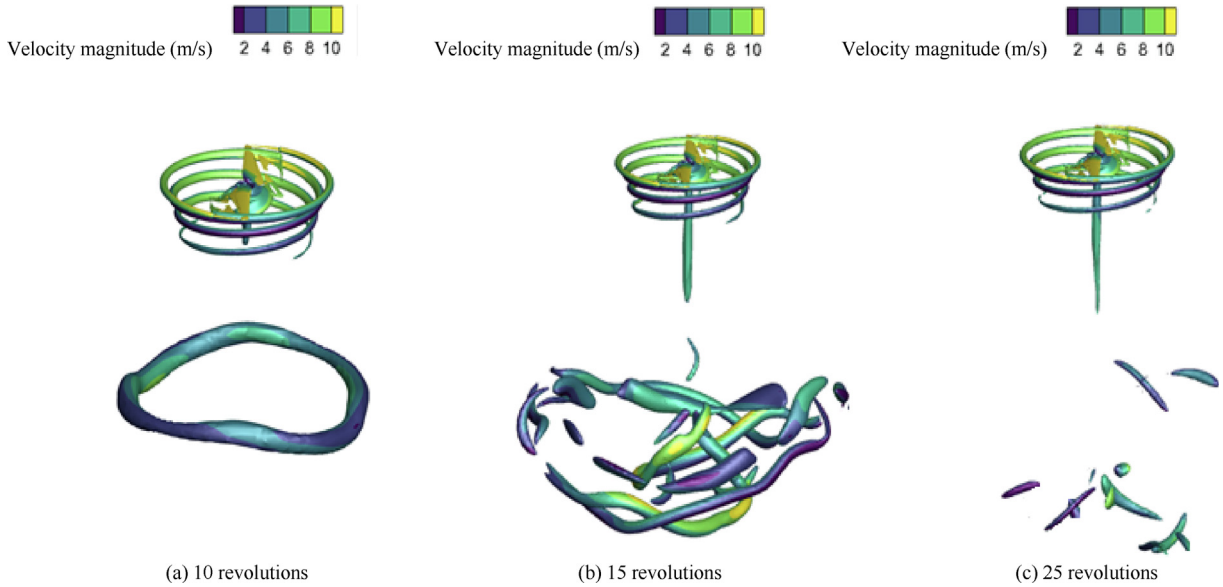


Fig. 3 Wake development after 10, 15, 25 revolutions showing velocity magnitude on isosurface using overset grid method.

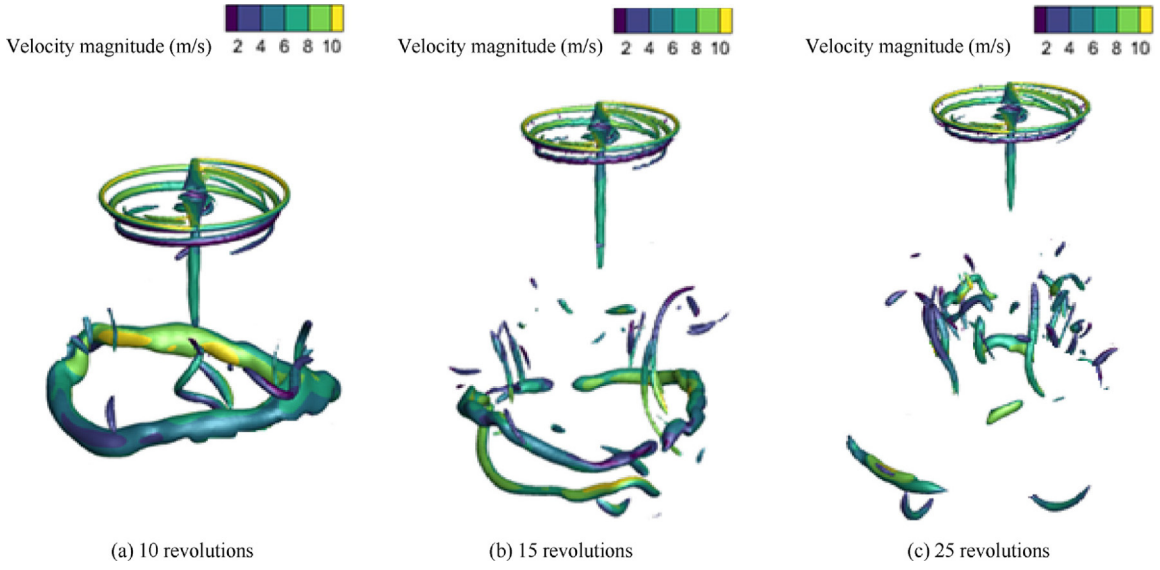


Fig. 4 Wake development after 10, 15, 25 revolutions showing velocity magnitude on isosurface using VBM.

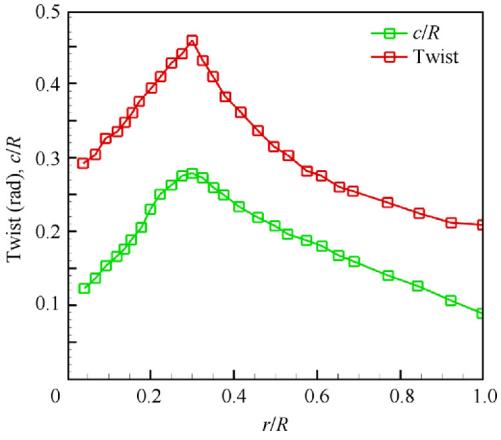


Fig. 5 Twist and chord distribution of DJI Phantom 2 rotor blades.

Table 2 DJI Phantom 2 parameters.

Parameter	Value
Diameter (m)	0.24
Number of blades	2
Solidity	0.12
RPM (r/min)	4860
Tip Mach number	0.18
Freestream velocity (m/s)	0
Reynolds number	6.2×10^4

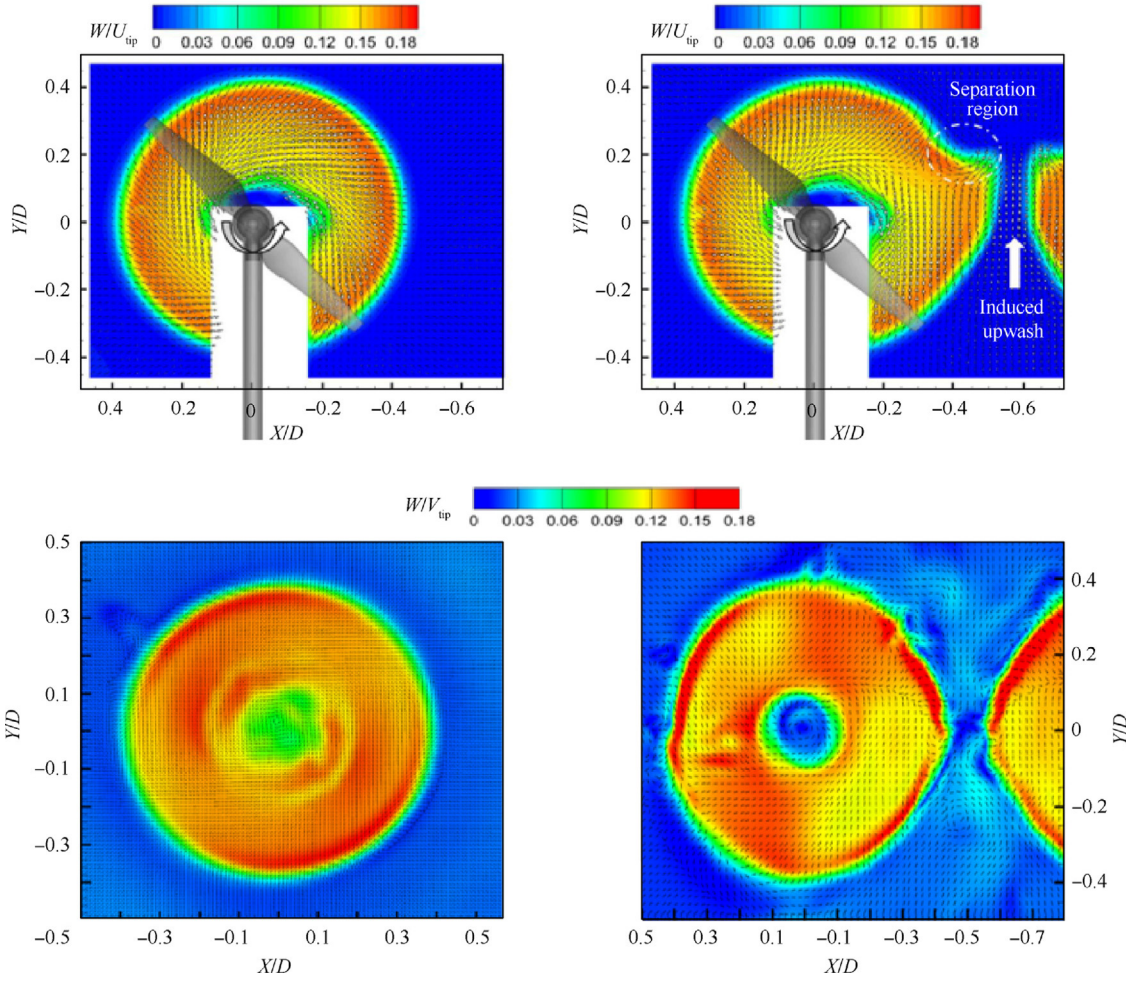


Fig. 6 Comparison of experimental results (top) and present analysis simulation results (bottom) for ensemble average velocity field of single rotor (left) and twin rotor (right) at $X/D = 0.1$.

2.1. Overset grid method

Overset grid method is considered as one of the most widely used meshing methods for rotor analysis[33–35] due to its great accuracy for steady and unsteady low analysis. An overset grid model comprises a fitted grid near a body, placed above a simpler underlying mesh, having the region of underlying mesh blanked out. Fig. 1 shows a typical overset grid model. Interface conditions are employed at the overset mesh boundaries in order to couple the solution between them. A drawback associated with this method is the involved computational cost which makes it not preferable for multirotor studies.

2.2. Virtual blade method

Virtual Blade Method (VBM) is a typical implementation of Blade Element Momentum theory (BEM) developed by Zori and Rajagopalan[36] for helicopter rotors application. It does not require actual rotor blade geometry or blade mesh, but simulates rotating blades effect on fluid through a body force acting inside a disk of fluid having an area equal to the swept area of actual rotor disk. Body force value of time averaged over a cycle from blade element method[37] calculated forces. It requires tables of lift coefficient and drag coefficient of selected airfoil at operating Mach numbers and angle of attacks along with geometric properties of the blades to replicate the flow effects of actual blades.

In BEM, a rotor blade is composed of several small segments or control points from root to tip of the blade. The drag and lift forces on each section are computed from 2D based on chord length, lift and drag coefficient of each segment, angle of attack, and airfoil type. Inlet boundary freestream velocity serves as the initial value for calculating angle of attack of each control point along the blade as shown in Eq. (3).

$$\alpha = \tan^{-1}\frac{u}{R\omega} \tag{3}$$

where α is angle of attack, u is streamwise velocity, R is rotor radius and ω is rotational velocity of rotor blades. Based on the values of angle of attack, lift and drag coefficients are interpolated from the provided tables. Using this information, lift and drag forces of each blade element is calculated by employing the formula in Eq. (4):

$$f_{L,D} = C_{L,D}(\alpha, Ma, Re) \cdot c_{r/R} \cdot \frac{\rho V_{\text{tot.}}^2}{2} \tag{4}$$

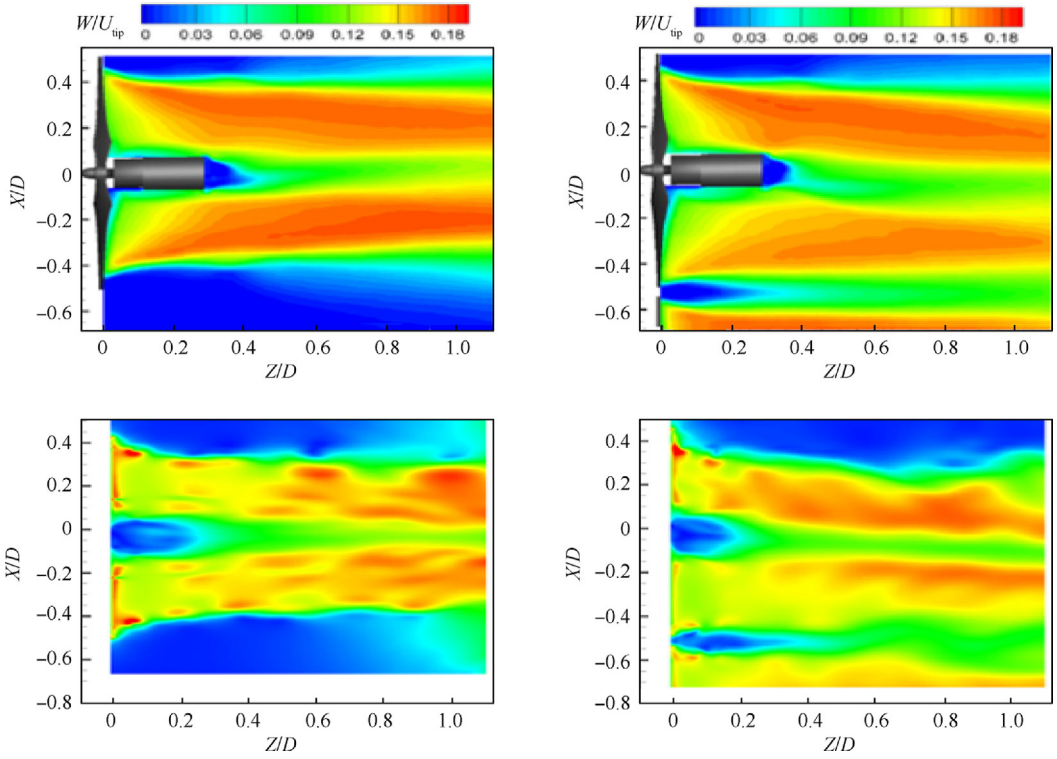


Fig. 7 Comparison of ensemble-average velocity experimental results (top) for isolated rotor and side by side rotor cases and for VBM simulation results (bottom).

where $f_{L,D}$ are lift and drag forces, $C_{L,D}$ are lift and drag coefficients, $c_{r/R}$ is chord length, ρ is flow density and $V_{\text{tot.}}$ is the flow velocity relative to blade. A typical VBM mesh is demonstrated in Fig. 2.

2.3. Comparison of overset grid method and VBM for rotor model under analysis

Overset grid method requires a geometry, therefore, CATIA V5 is used for that purpose. Structured hexahedral mesh is generated in ICEM CFD. The details of geometric model are mentioned in the next section. The mesh generated on blades is made fine near the tips in order to capture vortices. The first wall distance was kept small enough to keep y^+ values below 1. Value of thrust obtained in hover at 7000 r/min for isolated rotor model is 2.75 N using overset grid method.

Similar analysis is carried out employing VBM. Grid independence study is carried out to get the suitable accuracy comparable with overset grid method and reasonable mesh size. Three meshes were prepared and their corresponding thrust values were compared with value obtained using overset method. The value obtained from overset grid method is kept as standard to compare results of different VBM meshes, since overset grid method is quite famous for its accuracy. A detailed comparison of the methods along with computational cost involved is shown in Table 1.

Medium size mesh characteristics employing virtual blade method is selected for the further analysis of different octocopter configurations in hover and forward flight. A cylindrical region for each rotor disc with $0.015R$ disc density and two background grids on complete octocopter are utilized herein.

The fine background grid is $0.3R$ above and $3R$ below the octocopter and $5R$ from octocopter center in radial direction with $0.04R$ downwash density. The coarse background grid is $8R$ below, $1.5R$ above and $5R$ in sideways from octocopter center. For employing these mesh characteristics on complete octocopter in hover and forward flight cases, 26.72 million cells were used for each case utilizing 20.55 h/rev. The difference in required grid cells become very significant for multirotor study, as overset mesh method need grid cells for blades which is not required for VBM. A similar octocopter analysis with body fitted grid will require more than a billion cells making the analysis a daunting task due to the associated computa-

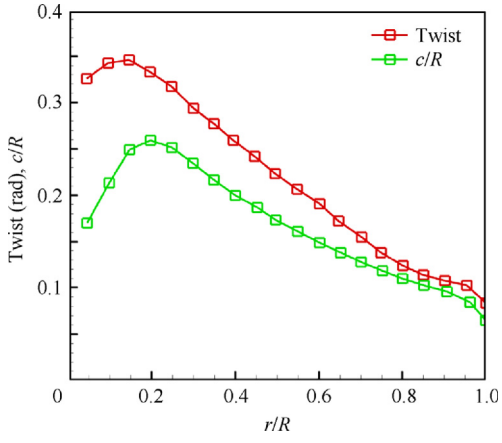


Fig. 8 Twist and chord distribution along the span of rotor blades.

Table 3 Parameters for octocopter configurations under analysis.

Parameter	Value
Number of blades	2
Number of rotors	8
Airfoil selected	NACA2411
Mean aerodynamic chord	0.01 m
Rotor diameter (Normal)	0.24 m (9.45 in)
Spacing from vehicle center to center of each rotor	0.35 m(13.78 in)
Vertical spacing in coaxial and hybrid configuration rotor layers	$0.5R$
Bigger rotor diameter	0.42 m(16.53 in)
Hover RPM	7000 r/min (733 rad/s)
Normal rotor tip Mach number in hover	0.256
Bigger rotor tip Mach number in hover	0.448
Octocopter angle of attack in forward flight	−7.7°
Inlet velocity for forward flight	15 m/s
Steady forward flight RPM	6000 r/min (628.3 rad/s)
Quasi-steady forward flight fore rotor RPM	5250 r/min
Quasi-steady forward flight aft rotor RPM	6750 r/min
Net difference of fore rotors and aft rotors in quasi-forward flight	Aft rotors with 25% more RPM than fore rotors
Total revolutions for each case	25
Time step size	0.5°
No. of iterations for 1 revolution	7200

tional cost. Both methods predict strong starting vortex after 10 revolutions which starts to weaken after 15 revolutions and disappears after 25 revolutions. This confirms that 25 revolutions can be considered enough for achieving stable results as shown in Fig. 3 and Fig. 4.

2.4. Validation of numerical method

For the verification of selected VBM method, simulations were conducted and compared with available experimental data from studies of Zhou et al.[38] to confirm the accuracy of selected method for current analysis. DJI Phantom 2 rotor models are used herein for the abovementioned purpose. The geometric and flow characteristics of the model are given in Fig. 5 and Table 2.

The comparison between experimental results and simulation results employing VBM in cases of a single isolated rotor as well as two side by side rotors experiencing rotor-to-rotor interaction is demonstrated in terms of the flow ensemble-average velocity at a plane located $0.1D$ downstream of the rotors in Fig. 6 and Fig. 7. The color map in the Fig. 6 represents axial velocity while the arrows show swirl velocity. The rotor tip speed is 61 m/s. Simulation result shows a bit of difference especially at the hub region. Due to this reason, the isolated rotor simulation converges to mean thrust coefficient value of 0.1005 and steady fluctuation value of 0.0011, while the experimental studies of Zhou et al.[38] and Ning [39] mention the experimental mean thrust coefficient of 0.1013 and steady fluctuation of 0.008. The error in result of mean thrust coefficient is about 0.5%, whereas the fluctuations are under predicted in the simulations. It can be assumed that the mounting pole/test stand had significant interactions resulting in fluctuations with the rotor which is not simulated in current VBM simulation. It can be observed that the wake structure is circular in isolated rotor case obtained through experiment as well as VBM simulation but it becomes conic in the region having two rotor tip in proximity of each other showing the interference effects on the shape of the wake. Since the two rotors are rotating in opposite direction, an upwash is generated between the rotors when the advancing rotor tips of two rotors come at the same location in proximity of each other. This phenomenon can be observed from both experiment and VBM simulation.

Fig. 7 demonstrates ensemble-averaged velocity as flow converted downstream of the single rotor and twin rotor cases for experimental and VBM simulation results. The results from experiments and VBM simulations are quite similar. From both studies, radial contraction of the induced flow (W component of velocity) is found showing that the high velocity region at rotor tip location ($X/D = 0.4$) converge to $X/D = 0.3$ near the root region as the flow moves downstream. The distribution of velocity is similar in both studies. Compared to isolated rotor case, the velocity field of twin rotor case in Fig. 7 is slightly dragged towards the neighboring rotor which could be the result of Coanda effect (i.e. phenomenon describing the flow to be attracted towards the nearby object). Due to this effect, the velocity in twin rotor case is observed to be lower than isolated rotor case in $X/D = -0.5$ to $X/D = -0.2$ region while the velocity distribution is almost the same on both sides of the isolated rotor.

3. Results and analysis

3.1. Geometric models and flow parameters

For analysis using VBM, tables of lift coefficient and drag coefficient against different angles of attack and Mach numbers are generated through 2D airfoil simulations of NACA2411 airfoil in Star-CCM+. NACA2411 is selected based on its high lift-to-drag ratios and previous use in octocopter studies. The rotor model selected for octocopter configuration analysis have the following geometric and flow

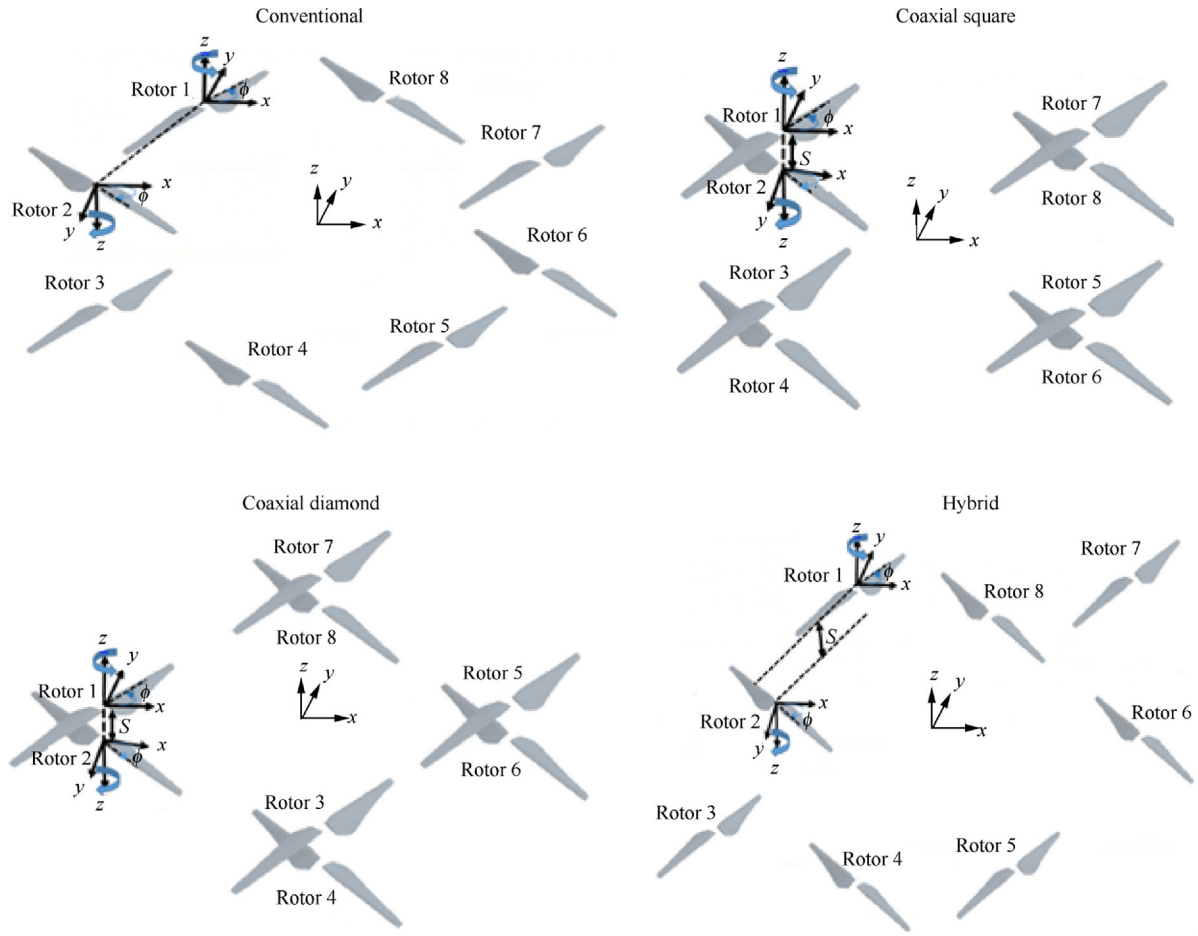


Fig. 9 Arrangement of rotors in different octocopter configurations.

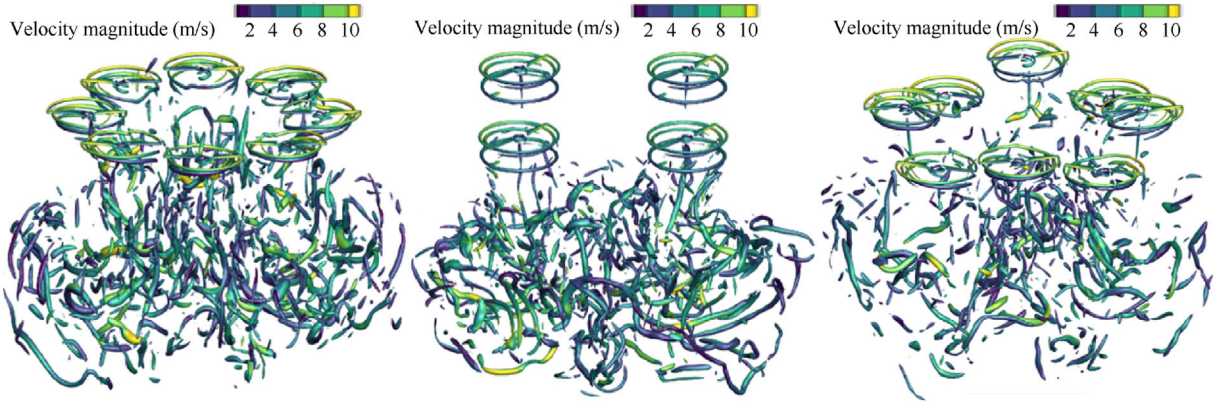


Fig. 10 Isosurface of velocity magnitude for octocopter configurations.

characteristics shown in Fig. 8 and Table 3. Rotation direction has a crucial effect on the stability characteristics of the octocopters and yaw trim becomes a must in order for the octocopter to avoid undesirable yaw.

To cope with this phenomenon, four rotors of each configuration are rotating counter-clockwise and four rotate clockwise. In conventional and hybrid configurations, every counter-clockwise rotating rotor is placed next to a clockwise rotating rotor, whereas the upper layer of rotors rotate counter-clockwise and lower layer rotors rotate clockwise in coaxial configu rations as demonstrated in Fig. 9.

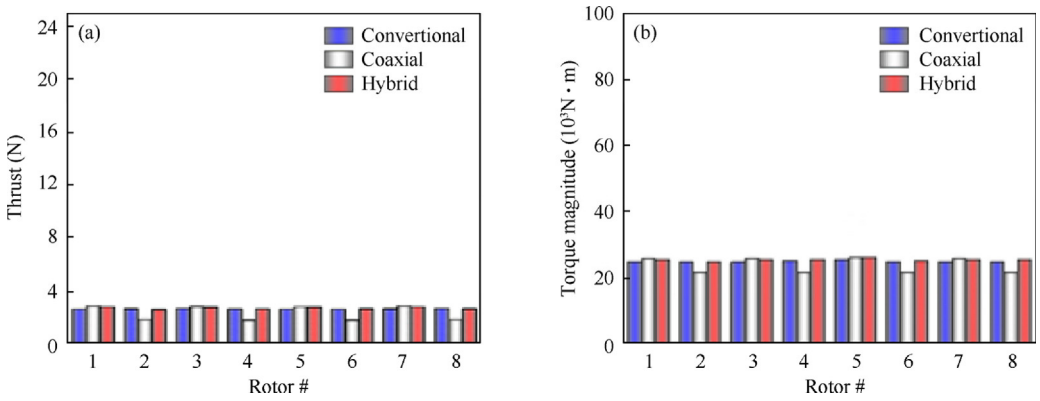


Fig. 11 Magnitude of (a) thrust and (b) torque produced by rotors of different octocopter configurations in hover.

Table 4 FM for different octocopter configurations in hover.

Type of configuration	$FM_{combined}$ value
Conventional	0.3156
Coaxial	0.2825
Hybrid	0.3219

3.2. Hover analysis

Hover flight analysis is carried out at 7000 r/min for all configurations. Here different configurations are compared to study how rotor-to-rotor interaction effects the thrust and torque magnitude of each rotor.

3.2.1. Rotor-to-rotor interaction in different configurations

Three major configurations are made part of the study for hover analysis i.e. conventional octocopter, coaxial octocopter and hybrid octocopter. Each case is run for 20 revolutions throughout the study. Vertical distance between two layers of coaxial and hybrid configuration is $0.5R$. Here R represents rotor radius of 120 mm. Rotor wake and rotor-to-rotor interaction is observed through isosurface using Q criterion value of 20000 s^{-1} and thrust distribution along the azimuth for rotors are obtained to understand this phenomenon. Thrust and torque magnitude are used as the criterion for comparison of rotor aerodynamic performance.

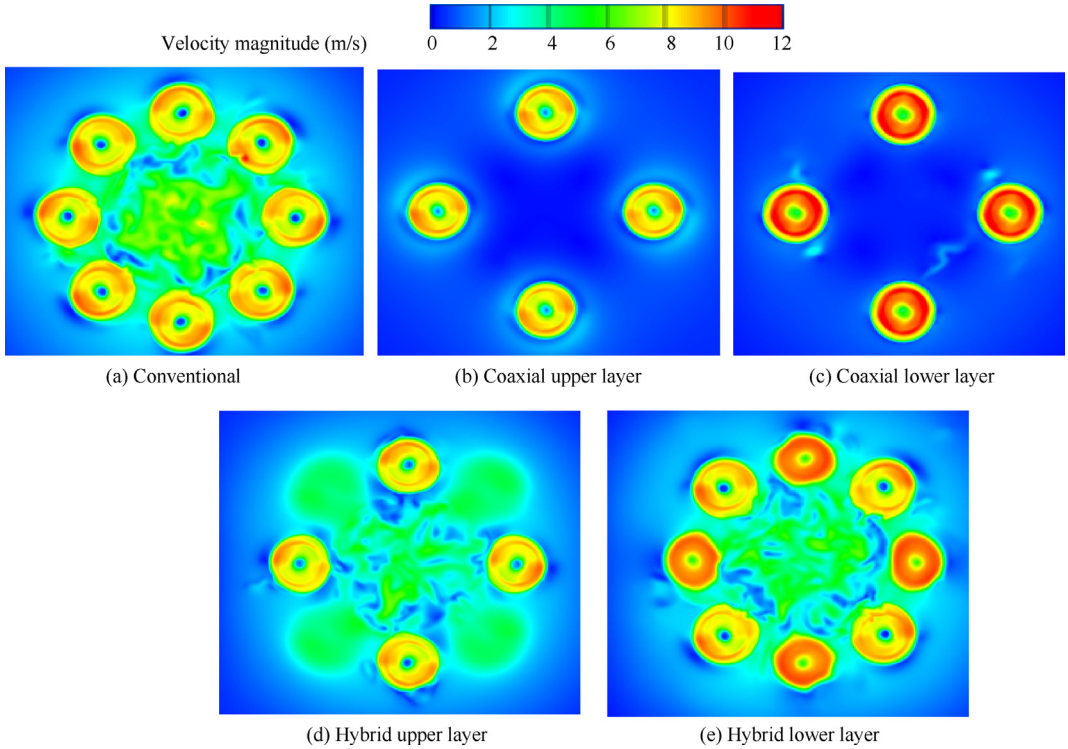


Fig. 12 Velocity magnitude contours for octocopter rotors in (a) conventional (b) coaxial upper layer (c) coaxial lower layer (d) hybrid upper layer and (e) hybrid lower layer.

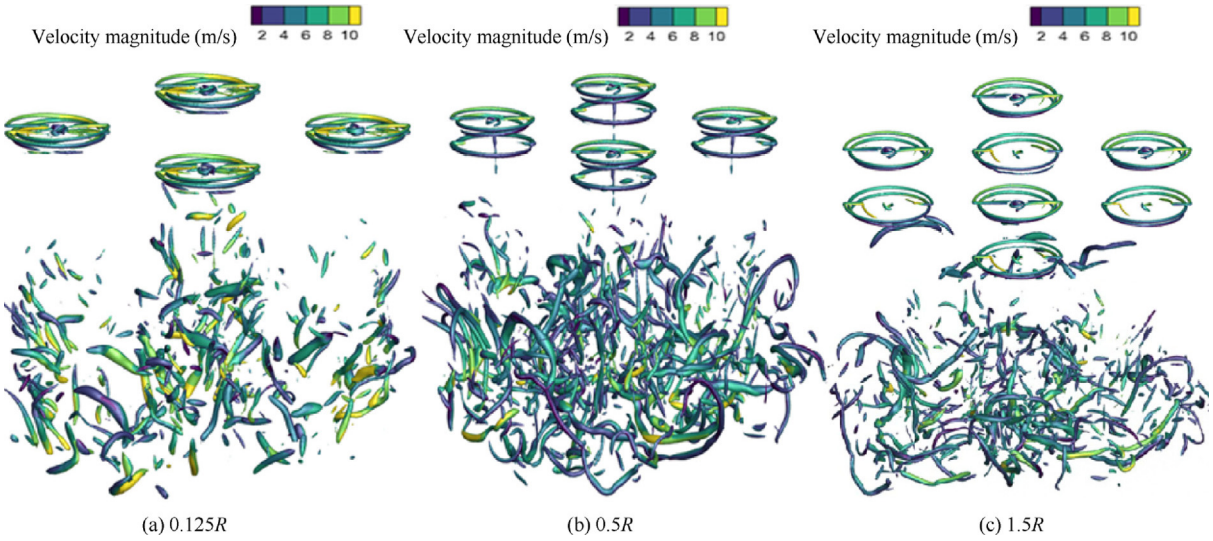


Fig. 13 Velocity magnitude isosurface for coaxial configuration with different vertical spacing.

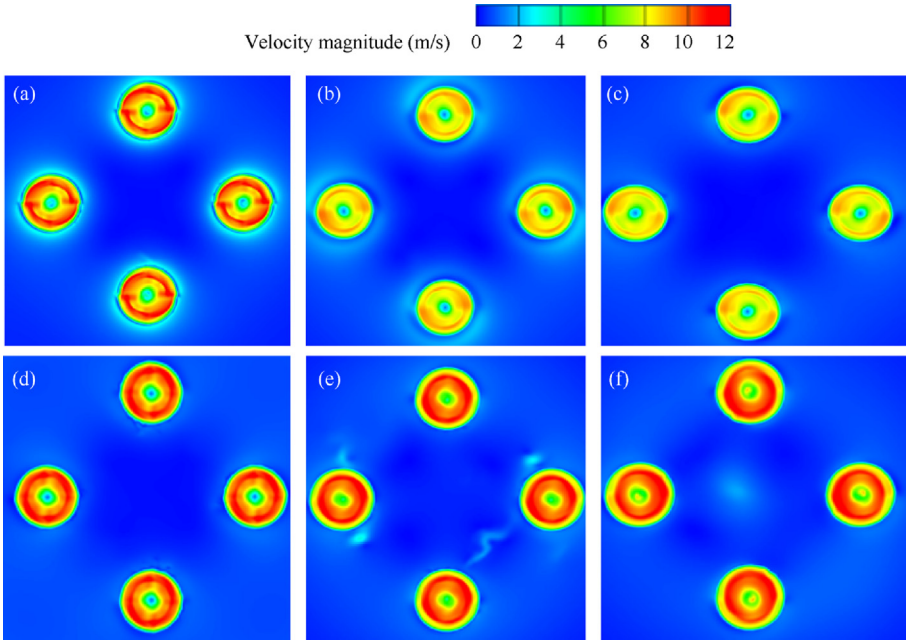


Fig. 14 Velocity magnitude for coaxial octocopter upper layer (a) 0.125R (b) 0.5R (c) 1.5R vertical spacing and lower layer (d) 0.125R (e) 0.5R and (f) 1.5R vertical spacing.

From the wake analysis in Fig. 10, it can be clearly observed that initially vortex structure is maintained for each configuration but as the wake moves downwards, there is vortex-vortex interaction which cause deformation of the wake structure and it splits into small segments in each configuration. In conventional configuration, it is also seen that a strong upwash is generated due to strong rotor-to-rotor interaction in the central region of the vehicle at fuselage location. This cause the wake to interact with fuselage also, which is not observed in coaxial and hybrid configurations. In other configurations, this phenomena can be observed downwards in the form of wake structure bent upwards between the rotors especially between coaxial rotor pairs. In conventional and hybrid configurations, wake deformation is more and wake interaction is wide spread but it is of greater strength in coaxial configration since lower layer of rotors is operating completely inside the wake of upper layer rotors. Thrust fluctuations further verify the mentioned statement. Thrust and torque magnitude for each rotor in Fig. 11 demonstrates a clear degradation in aerodynamic performance of each octocopter configuration. FM is depicted in Table 4 shows conventional and hybrid configurations having similar combined FM values which are significantly larger than FM value for coaxial configuration.

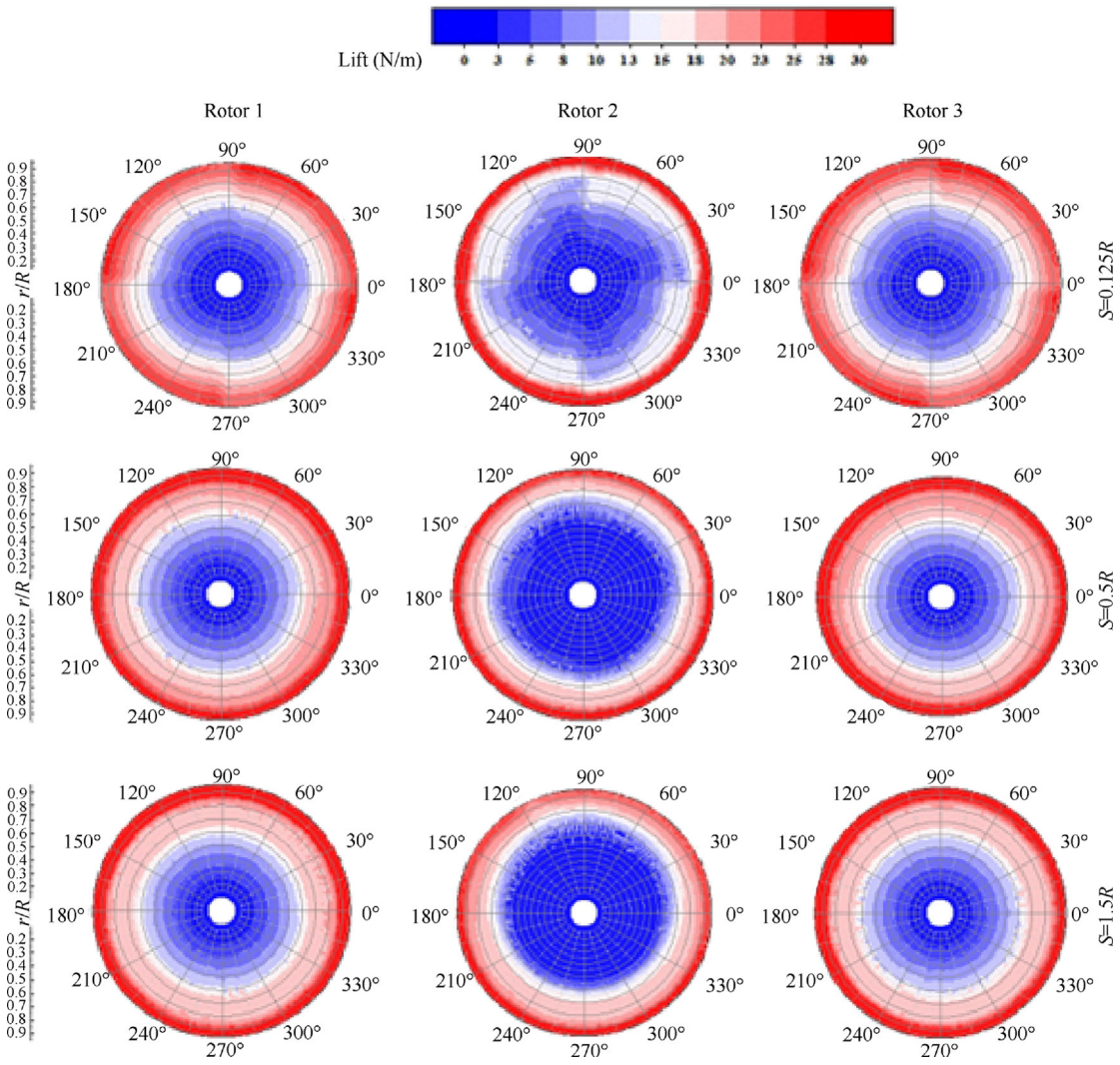


Fig. 15 Sectional lift distribution for hovering coaxial octocopter with different vertical spacing.

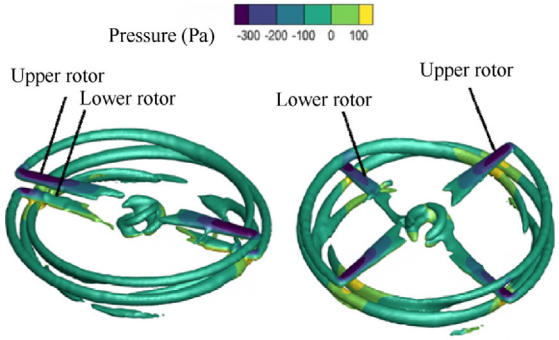


Fig. 16 Loading effect on upper surface of lower rotor at 0.125R vertical spacing showing changes in pressure.

All the rotors of conventional configuration are effected almost equally but not significantly due to proximity of each rotor tip with neighboring rotor tip. Fig. 12 demonstrate negligible wake interaction showing same velocity at each rotor and wake of each rotor interact in the central region of the vehicle not effecting the aerodynamic performance of the rotors. Similar thrust values for each rotor of conventional configuration shown in Fig. 11 also confirms it. Hybrid configuration is a hybrid of conventional and coaxial configuration with its clockwise rotating four rotors moved vertical down from conventional configuration.

It is designed with the aim to reduce tip-to-tip interaction as well as providing the opportunity to expand rotor diameter in same confined space for high thrust requirements. The lower layer of hybrid configuration is effected by the wake from upper layer but not as much as the lower layer of coaxial configuration, since hybrid configuration rotors don't operate completely below the upper layer. The velocity distribution on lower layer of hybrid configuration shows almost no difference compared to upper layer. The lower layer of coaxial rotor is effected significantly shown by a significant increase in velocity in center region of rotors in lower layer. Total thrust and torque magnitude values of conventional and hybrid configurations are also similar but coaxial configuration shows significant drop in total thrust of 11.89% and 14.22% as compared to conventional and hybrid configurations. Torque magnitude

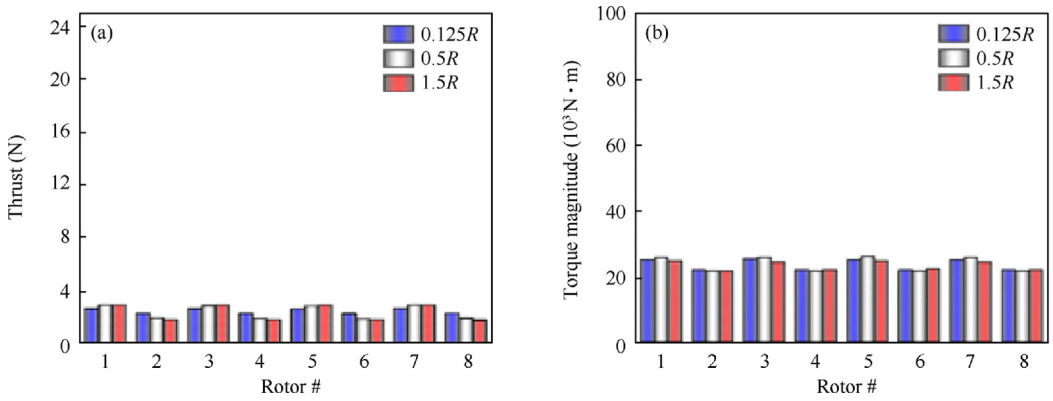


Fig. 17 Magnitude of (a) thrust and (b) torque for hovering coaxial octocopter with different vertical spacing.

Table 5 FM for coaxial octocopter with different vertical spacing.

Vertical spacing (S)	$FM_{combined}$ value
0.125R	0.2931
0.5R	0.2826
1.5R	0.2829

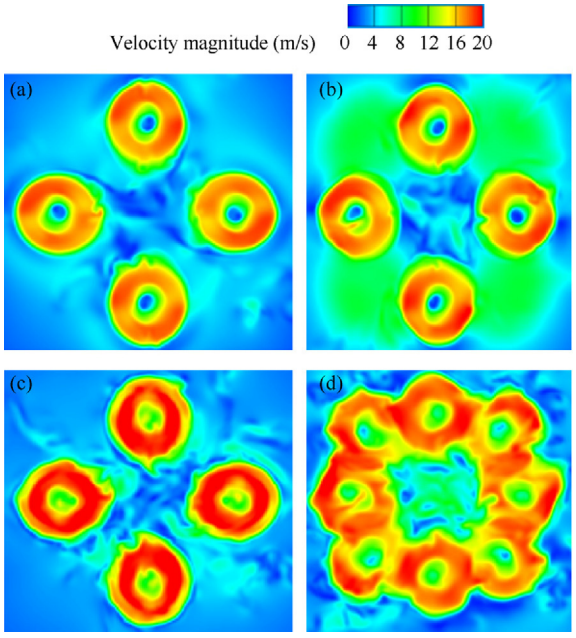


Fig. 18 Velocity magnitude at upper rotor layer of (a) coaxial (b) hybrid configurations and at lower rotor layer (c) coaxial and (d) hybrid configurations.

also shows a similar trend which results in instability of the vehicle due to rotor-to-rotor interaction and consumption of more power.

3.2.2. Effect of vertical spacing in coaxial octocopters on rotor-to-rotor interaction

Vertical spacing (S) between the upper and lower layers of rotors could be a crucial factor in design of these configurations, therefore, a comparative study is carried out to study the effect of variation in vertical spacing on rotor-to-rotor interaction and subsequently on thrust and torque magnitude of the vehicle. Hybrid configuration showed no significant influence of vertical spacing on thrust and torque magnitude as far as the spacing it not reduced to a level that it starts behaving like a conventional configuration. Coaxial configuration is studied for vertical spacing of 0.125R, 0.5R and 1.5R between the two rotor layers. The velocity isosurface, ensemble-average velocity and distribution of sectional lift along the azimuth of rotor are demonstrated in Fig. 13, Fig. 14 and Fig. 15. Thrust distribution is illustrated only for rotors 1, 2, 3 because similar characteristics are shown by the rest of the rotors also. It is observed that the degradation of thrust of lower rotor layer in hover is at the central region of the rotors, since the upper rotor wake contracts and goes into the central region of lower rotor and tip region is not effected much. This effect is quite significant from rotor root to 0.8R span.

Fig. 14 shows the ensemble-average velocity 0.1D below the rotor layer for each vertical spacing case. Comparing upper layers show only 0.125R effected at locations where it comes in proximity of lower rotor blades. The lower layers show increase in downwash at the center region with increase in vertical spacing leading to lower thrust. Similar results are found in the studies of Lakshminarayan [40] and Park[41] et al.

From Fig. 13, it is observed that very less wake interaction in coaxial rotors having vertical spacing of 0.125R and wake structure is not destroyed much for this case. This shows that at very less vertical distance, wake interaction is not significant in coaxial rotors. In fact, at very small vertical spacing, wake interference is dominated by loading effect.

When two coaxial rotors are very close, they effect pressure distribution on the surfaces facing each other. The lower surface of upper layer of rotors enter low pressure region due to low pressure on upper surface of lower layer of rotors leading to reduction of pressure gradient on surfaces of upper layer of rotors. Similarly, the upper surface of lower layer of rotors enter high pressure region due to high pressure on lower surface of upper layer of rotors leading to increase in pressure of upper surface of lower layer of rotor. This subsequently decrease pressure gradient on the surfaces of lower layer of rotors. Reduction in pressure gradient on both upper and lower layer rotors takes place leading to reduction in thrust of both upper layer of rotors and lower layer of rotors. Fig. 16 shows increase in pressure of upper surface of lower

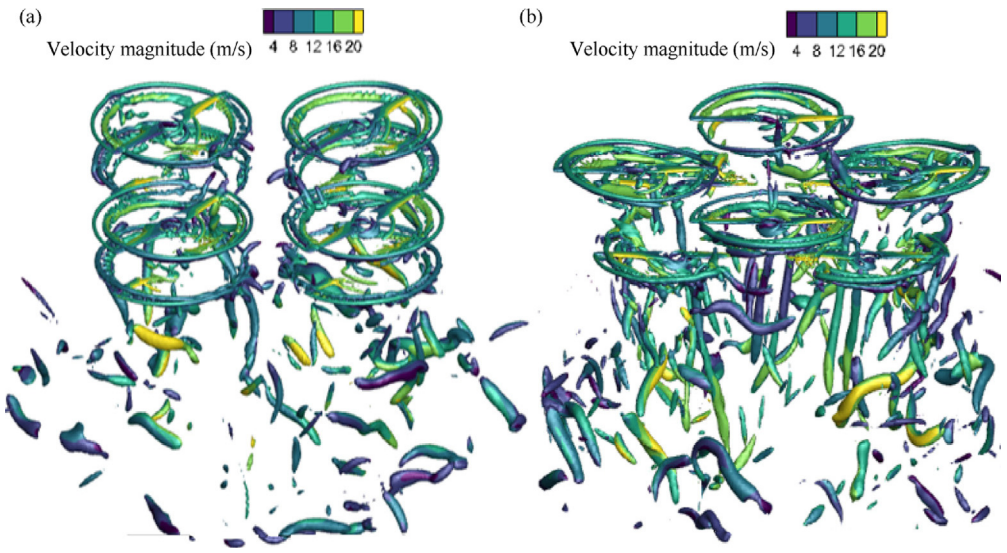


Fig. 19 Isosurface of velocity magnitude at Q criterion 100000 s^{-2} for (a) coaxial and (b) hybrid octocopters with bigger diameter.

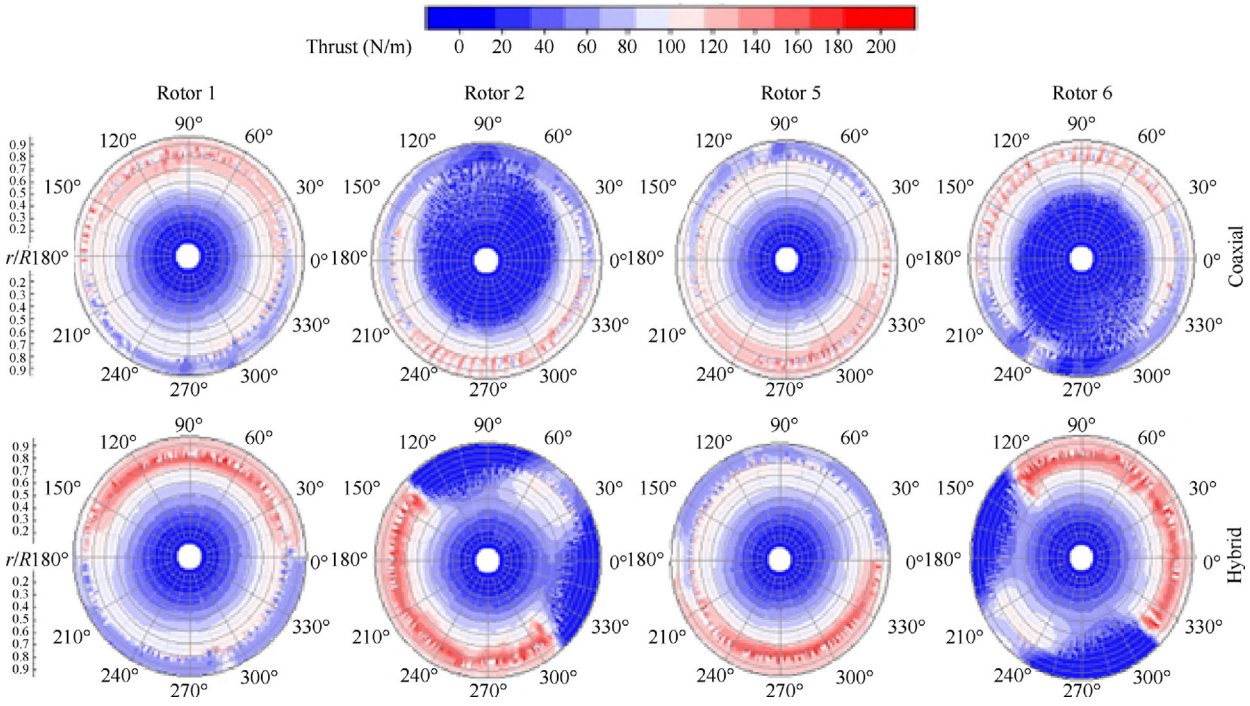


Fig. 20 Thrust distribution along azimuth of hovering octocopters for coaxial and hybrid configurations with bigger diameter rotors.

rotor blades when then come exactly below the upper rotor and pressure of lower rotor blades become same as upper rotor blades when then are not overlapping. This phenomenon is also demonstrated in Fig. 15 where the blade of upper rotor layer comes in contact four times in a revolution, leading to degradation of thrust after every 90° of rotation. It is also confirmed by available literature.[42] Average thrust values shown in Fig. 17 for 0.125*R* case further elaborate the aforementioned phenomenon. As rotor spacing is increased, loading effect diminishes and wake interaction effects become significant. The thrust of top rotor starts to increase and it decreases for lower rotor but total thrust is not influenced significantly. Further increase in vertical distance does not change thrust and torque magnitude values much. Table 5 shows that FM value is best for 0.125*R* spacing but there is no significant change in combined FM with increase in vertical spacing.

3.2.3. Comparison of coaxial and hybrid configurations with bigger diameter rotors

It is worth mentioning that coaxial and hybrid configurations provide the opportunity to expand the diameter of rotors within the confined space for high thrust requirement missions. A comparative study of coaxial and hybrid octocopters with

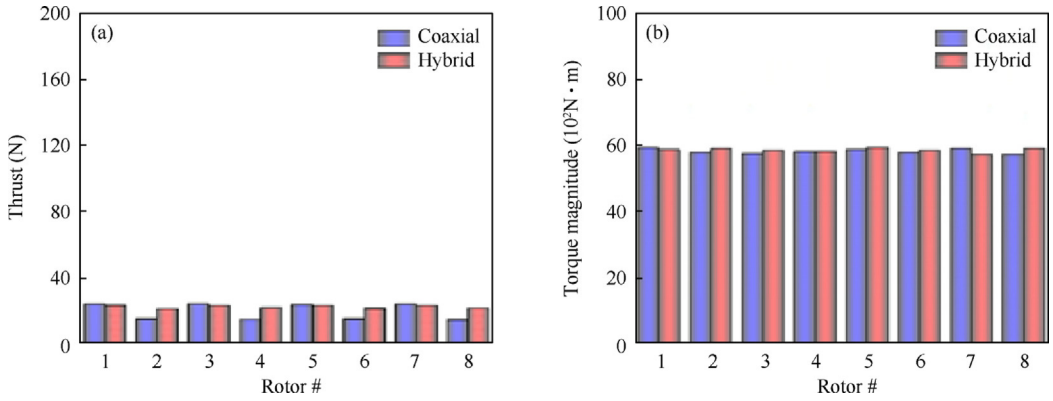


Fig. 21 Magnitude of (a) thrust and (b) torque produced by bigger diameter rotors of coaxial and hybrid configurations in hover.

Table 6 FM for coaxial and hybrid configurations with bigger rotors.

Type of configuration	$FM_{combined}$ value
Coaxial	0.3066
Hybrid	0.3673

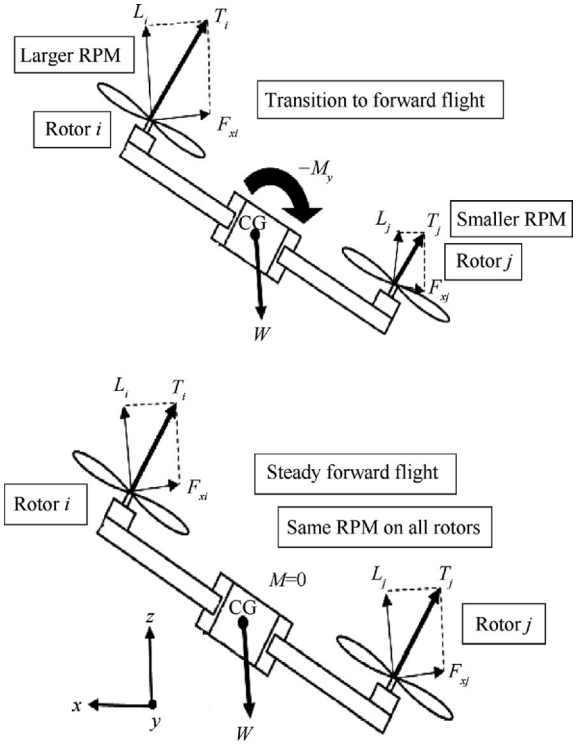


Fig. 22 Forces and moment on a multirotor in transitional and steady forward flight case.

bigger diameter rotors is presented herein. The diameter for each rotor is 420 mm. The velocity isosurface and thrust distribution along with average thrust and torque magnitude values for each rotor are demonstrated. It is observed that with bigger rotors, the interaction is more significant and tip-to-tip interaction also becomes quite apparent which was not significant in normal size rotors.

Fig. 18 shows the ensemble-average velocity for rotor layer of different octocopter configurations, while Fig. 19 shows rotor-wake interaction for this case. Fig. 20 shows that the effect of upper rotor layer wake on lower layer rotors is significant from rotor root to almost $0.8R$ in coaxial configuration, whereas it is significant near the tip regions in hybrid configuration since each lower rotor in hybrid configuration is partially overlapping two upper layer rotors. Although hybrid configuration lower layer rotor is effected by two upper layer rotors, but still degradation of aerodynamic performance is less than corresponding coaxial configuration rotor. Rotors 1, 2, 5, 6 are used for analysis of thrust distribution along the azimuth since these four rotors can demonstrate the characteristics of interaction in each rotor with other rotors having similar characteristics with just different direction of interaction based on respective location in the vehicle. Upwash can be observed moving up at the center of the vehicle in hybrid configuration which is not so apparent in coaxial configuration. Since the upper layer rotate anticlockwise, so its tip to tip interaction with neighboring rotors is at $+180°$ to $+270°$ whereas lower layer rotate clockwise so its interaction is at $-180°$ to $360°$ ($0°$–$180°$) for same side rotors placed one above the other. Rotors 1, 2 are fore rotors on the right side of the vehicles whereas rotors 5, 6 are aft rotors present on left side of the vehicle.

With bigger diameter rotors, a significant difference of 15.61% can be seen between coaxial octocopter and hybrid octocopter from Fig. 21, whereas torque magnitude is quite similar. Table 6 showing FM for both configuration also confirms better aerodynamic performance of hybrid configuration with bigger rotors in hover. Fig. 18 shows that lower layer of coaxial configuration with big rotors experience strong downwash from the upper rotor layer which effects the central region the most and covering almost whole rotor, whereas the lower layer of hybrid configuration does show significant effect of downwash on large part of the rotors like coaxial configuration and only sides of the rotors in lower layer get interaction with wake from upper rotors reducing the aerodynamic performance of lower layer of hybrid configuration significantly less than that of lower layer of coaxial configuration.

This also shows that thrust performance of hybrid octocopters is better than coaxial octocopters for high thrust

Table 7 Isolated rotor characteristics at different RPM.

Rotor RPM (r/min)	Lift (N)	Forward force (N)	Torque magnitude (N·m)
5250	1.96	0.102	0.0159
6000	2.52	0.121	0.0165
6750	3.38	0.143	0.0221

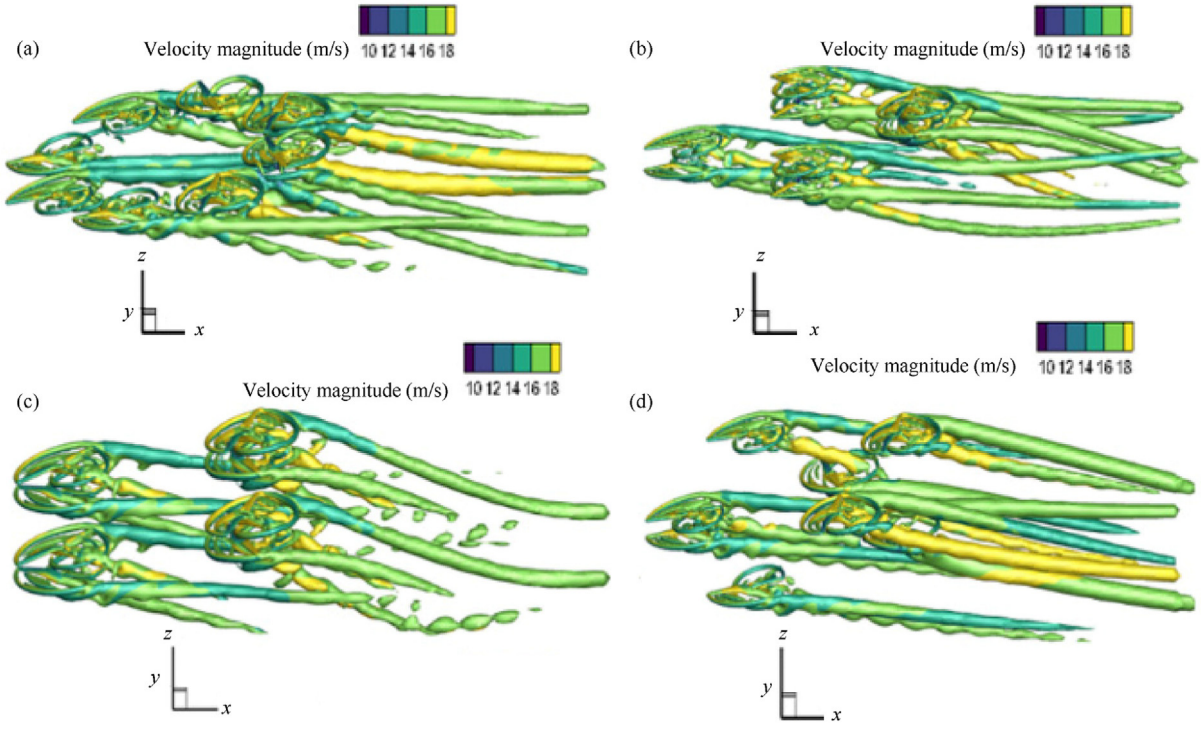


Fig. 23 Isosurface of velocity magnitude for quasi steady forward flight of (a) conventional (b) coaxial diamond (c) coaxial square and (d) hybrid octocopters.

requirements also with similar power consumption, since torque magnitude is associated with power consumption. It is clear from thrust values that the difference in total thrust is mainly due to lower layer of coaxial rotors operating exactly under the wake of upper layer rotors. Whereas, in hybrid rotors, there is a partial overlap. The effect of partial overlap can be clearly observed from the isosurface contour of hybrid configuration as well. The part of the vortex ring coming exactly below the upper rotors is diminished causing degradation and fluctuations of thrust in that region. The isosurface plot of coaxial configuration also show the central area of lower rotors diminished where a virtual blade was present demonstrating the fact that wake of upper rotors effect the central region of lower layer rotors herein.

3.3. Forward flight analysis

Two types of forward flight is considered herein i.e. quasi-steady forward flight and steady forward flight. Since transition from hover to forward flight of an octocopter is achieved by reduction in rpm of fore rotors and increase in rpm of aft rotors so that the vehicle can tilt forward, this type of motion with different rotation speeds of fore rotors and aft rotors with approximation of insignificant difference in thrust on different rotors is called as quasi-steady forward flight. After achieving the desired tilt angle, the octocopter needs to increase the rpm of fore rotors and reduce rpm of aft rotors in order to achieve same rpm on all rotors so that it does not tilt further and it can move forward horizontally. This type of forward flight has same rpm on all rotors of an octocopter. It is called as steady forward flight. Although very less literature in available on forward flight of a multirotor, most of it considers steady forward flight case. All forward flight simulations for present study are carried out at angle of attack of $-7.7°$. The distance between two layers of rotors in each configuration is kept $0.5R$. Radius and geometry of rotors is same as used in hover analysis. Fig. 22 shows the phenomenon of transition from hover to forward flight and steady forward flight.

Simulations are performed first on an isolated rotor under same flow conditions as those which will be employed on dif-

ferent octocopter configurations in different types of forward flight in order to know performance of isolated rotor which could be compared with rotors of octocopter under rotor-to-rotor interaction. The values for forward thrust and torque magnitude in different flow conditions employing present methodology are shown in Table 7.

Analysis of an isolated rotor at fixed tilt angle and different rotation speeds demonstrate the behavior of lift and forward thrust generated by the rotor. It is seen that increase in rpm increase thrust and torque generated by the rotor which subsequently increase the upward lift and forward force (F_x) components.

3.3.1. Quasi-steady forward flight

Quasi steady forward flight an octocopter is simulated by reducing the rpm of fore rotors by adjusting forerotor and aft rotor rpm to achieve 25% net difference which is appropriate for quasi-steady forward flight of such vehicles.[43] Applying the required rpm changes, fore rotors rotate at 5250 r/min and aft rotors rotate at 6750 r/min. Three major configurations are studied herein i.e. conventional, coaxial and hybrid octocopters. Coaxial octocopter can move forward in a diamond configuration, as well as square configuration. Therefore, both of these configurations are studied. The rotor position and rotation direction for each configuration is already depicted earlier.

It is worth mentioning that the four front rotors in conventional, hybrid and coaxial square octocopter configurations will have 5250 r/min and the four aft rotors have 6750 r/min but coaxial diamond configuration cannot use all the rotors for transition from hover to forward flight. Its side rotors stay with the same RPM i.e. 6000 r/min and it can only reduce rpm of two front rotors to 5250 r/min and increase rpm of two aft rotors to 6750 r/min. As coaxial diamond configuration uses only four rotors for achieving required forward tilt, its transition is slow, having yawing moments and higher thrust effort is required for the four rotors involved in achieving the tilt angle. All cases of quasi forward flight are analyzed at 15 m/s inlet velocity. The results for velocity isosurface at Q criterion of 16000 s^{-1} and respective lift and forward thrust generated by each configuration is as shown in Fig. 23 and Fig. 24 shows the lift and torque magnitude comparison for each rotor.

Observing the illustrated results show that lower layer of coaxial rotors have lower lift and forward thrust compared to upper layer rotors due to being operated inside the wake of upper layer rotors. It is also observed that the advancing blades of front two rotors in conventional configuration approach each other and generate an upwash increasing their lift. This upwash travel downstream with flow and increase lift of rotor 6 and rotor 7 present exactly behind. In coaxial square configuration, four rear rotors operate in the wake of four front rotors leading to a significant loss of performance of overall vehicle, whereas coaxial diamond configuration has two rear rotors operating inside the wake of two front rotors, showing lesser extent of overall performance deterioration. Conventional and hybrid configurations performance is similar to performance of isolated rotors operating in identical flow conditions whereas coaxial configuration show loss in performance. The vortices produced by rotors move in the flow direction and merge together to form bigger.

3.3.2. Steady forward flight

The second type of forward flight has same rpm on all rotors of the octocopters and they fly horizontally forward in a steady manner. Here angle of attack is −7.7° and inlet velocity is 15 m/s. The velocity isosurfaces for Q criterion value of 16000 s^{-2} is illustrated in Fig. 25 for each configuration.

Velocity distribution on each layer of octocopter configurations in steady forward flight are demonstrated in Fig. 26 to have a better understanding of rotor-to-rotor interaction. Thrust distribution along the azimuth of rotors 1,2,5,6 of every configuration in steady forward flight is given in Fig. 27. Rest of the rotors depicts similar aerodynamic characteristics. It is observed that the fore rotors wake degrades thrust distribution of aft rotors at angles between 30° to 180° and the upper layer

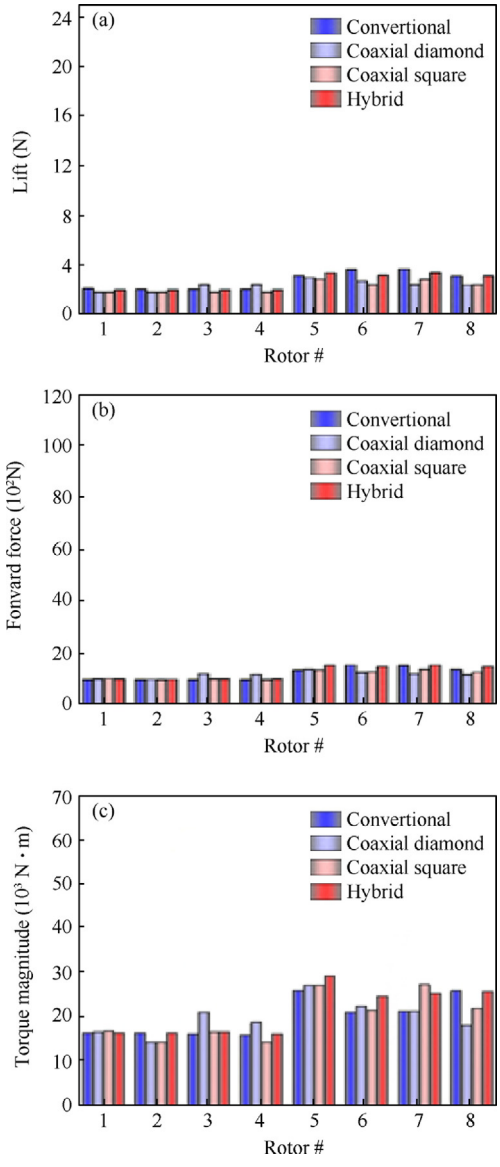


Fig. 24 Magnitude of (a) lift (b) forward force and (c) torque produced by rotors of different octocopter configurations in quasi-steady forward flight.

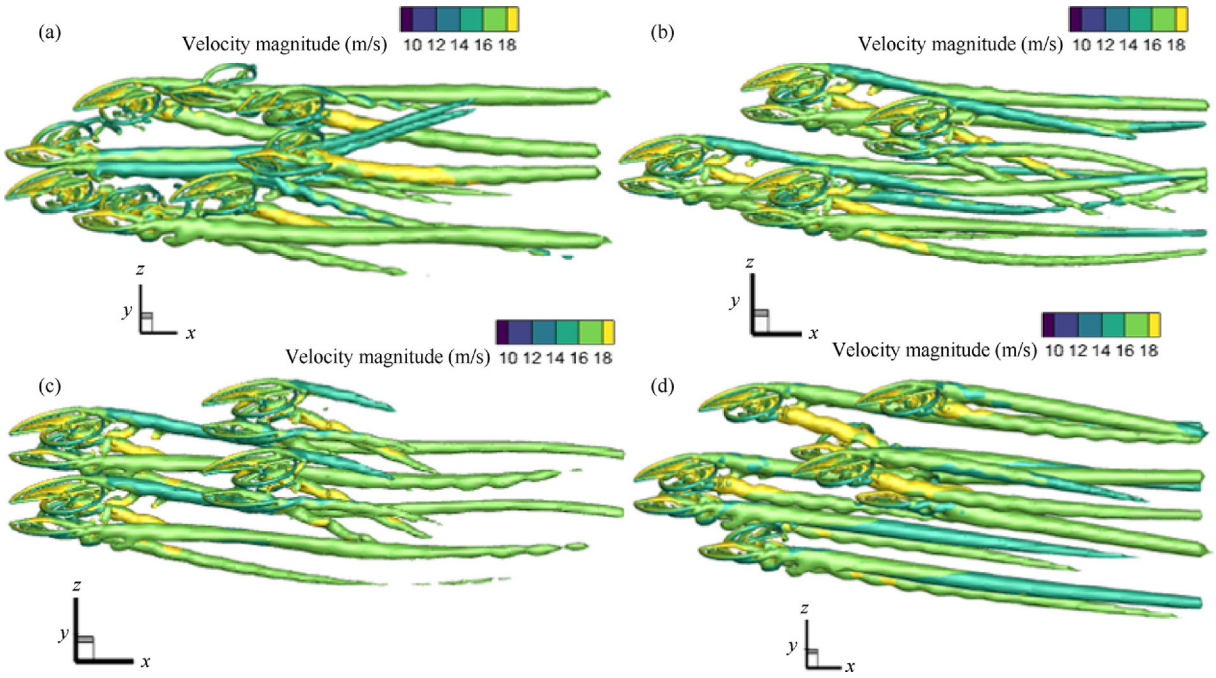


Fig. 25 Isosurface of velocity magnitude at Q criterion 16000 s^{-2} for (a) conventional (b) coaxial diamond (c) coaxial square and (d) hybrid octocopters in steady forward flight.

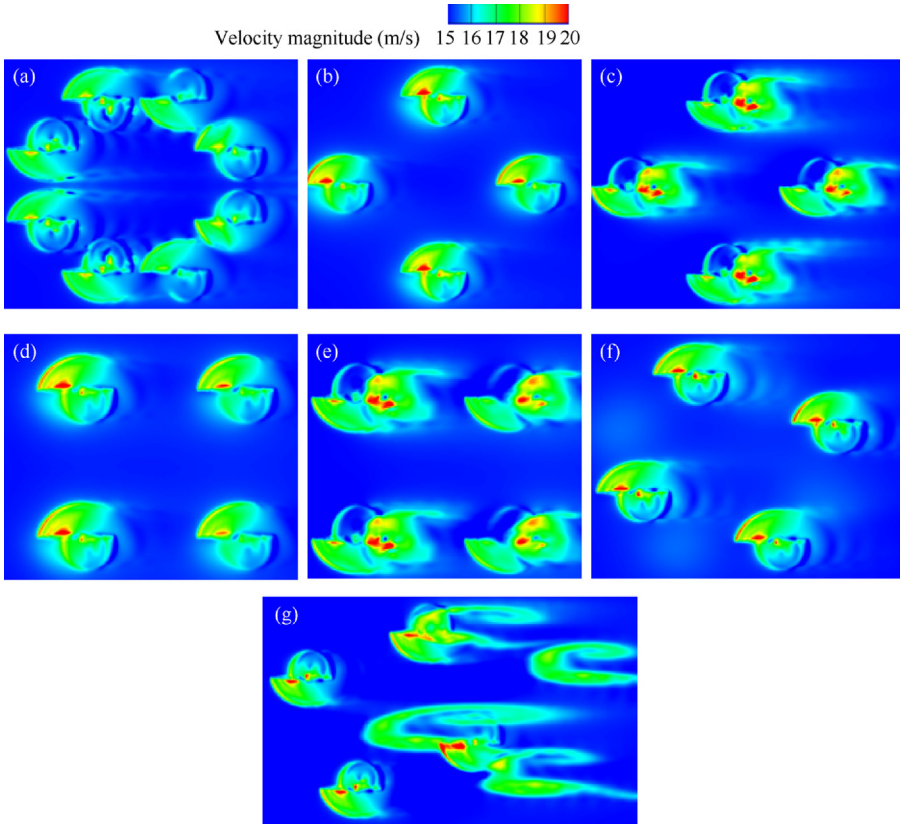


Fig. 26 Velocity magnitude on rotors of (a) conventional (b) coaxial diamond upper layer (c) coaxial diamond lower layer (d) coaxial square upper layer (e) coaxial square lower layer (f) hybrid upper layer (g) hybrid lower layer in steady forward flight.

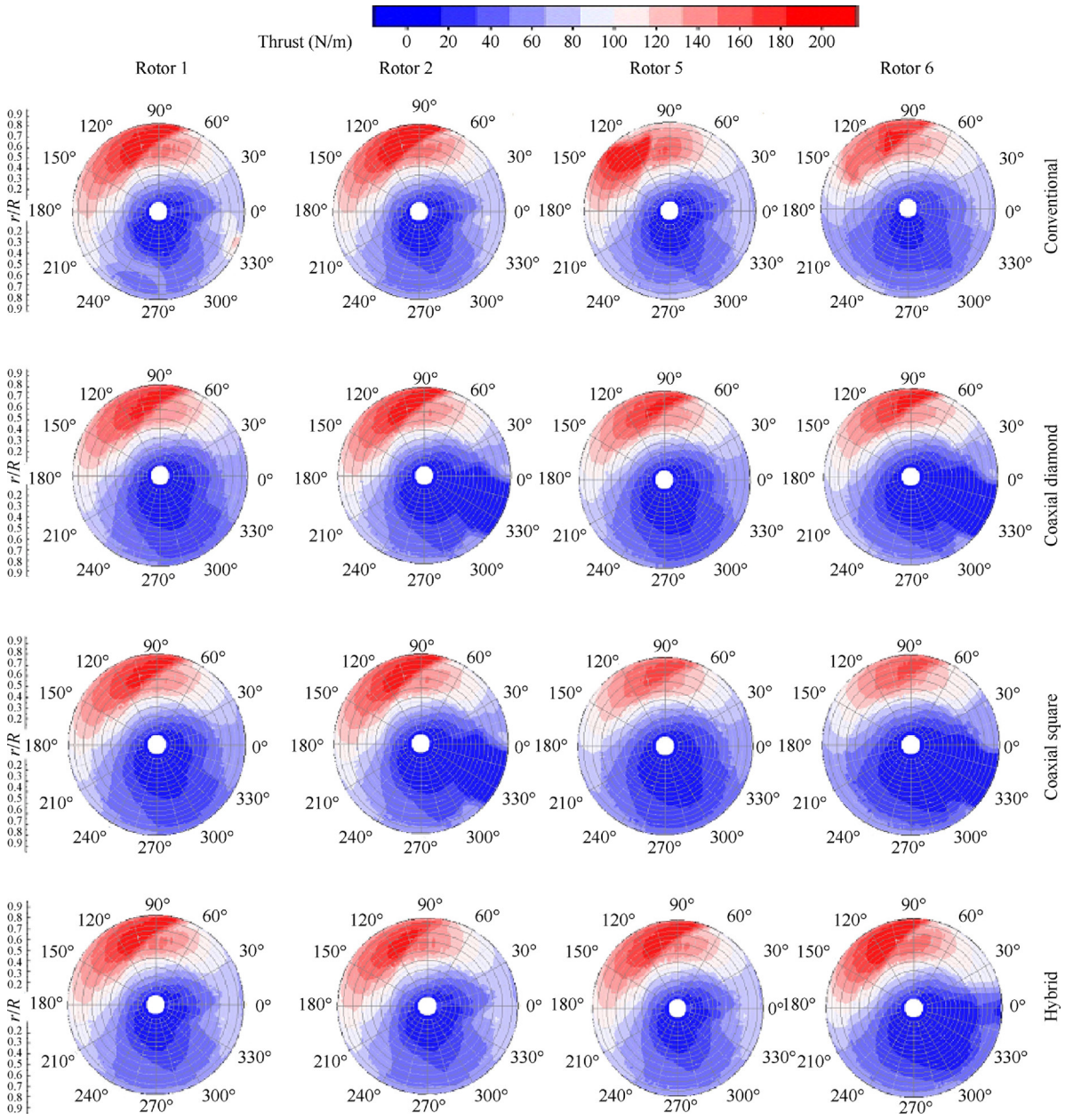


Fig. 27 Thrust distribution along rotor azimuth in steady forward flight for different configurations.

of rotor wake deteriorates the thrust of lower layer rotor at 270° to 360° azimuth of the rotor. The aft rotors present in lower layer are most effected, since they face the wake of fore rotors as well as upper rotor layer. The extent of performance degradation of aft rotors due to wake of fore rotors is comparatively less than the effect on lower rotor due to presence of a rotor above it. Tip-to-tip interaction is not so apparent. The lift, forward force and torque magnitude values in Fig. 28 clearly demonstrate that the front four rotors of conventional and hybrid configurations have more lift than isolated rotor but the forward force is not more than isolated rotor. The reason behind this phenomenon is the upwash created by rotors rotating in the opposite directions having advancing blade tips of two rotors near each other. When advancing blades of two side by side rotors come in proximity of each other, they generate upwash leading to increase in lift of both rotors. This phenomenon can be clearly observed from all velocity isosurface contours of conventional rotors in particular. A clear upwash can be observed between the front two rotors which moves downstream with the flow. This flow is more prominent in conventional configurations resulting in more lift increase than hybrid configuration because its rotors are placed side by side and very close to each other, whereas rotors of hybrid configuration are not so close but still a notable increase in lift in hybrid configuration is also observed. Comparing thrust of rear rotors of conventional configuration, it is seen that lift of rotor 6 and rotor 7 is more than rotor 5 and rotor 8. This is because the upwash generated by the front two rotors which travel downstream straight between rotor 6 and rotor 7 present exactly at the back of front two rotors, causing beneficial effect

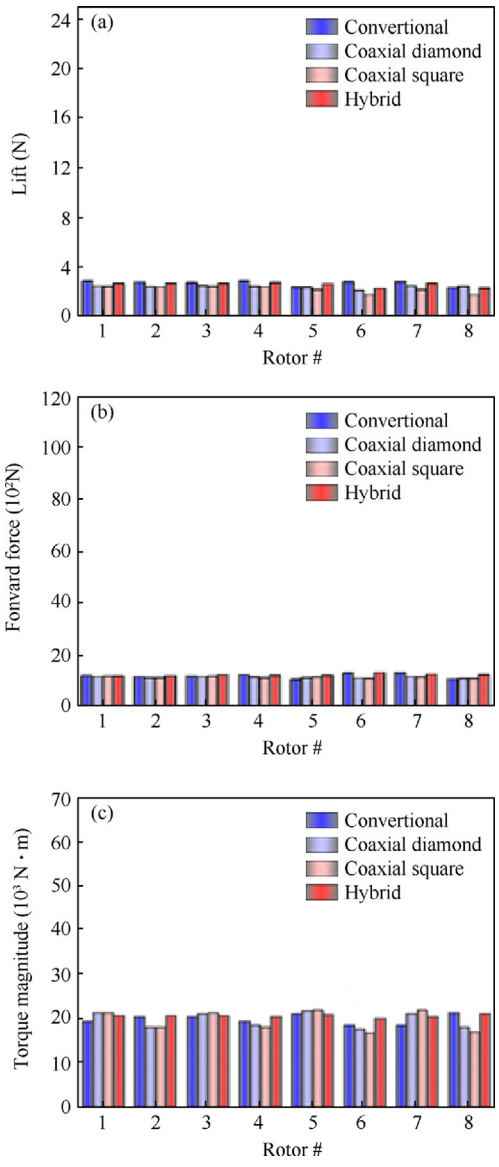


Fig. 28 Magnitude of (a) lift (b) forward force and (c) torque produced by rotors of different octocopter configurations in steady forward flight.

on these two rotors as well. Another important observation is degradation of rotor 6 and rotor 8 of hybrid configuration, whereas rotor 5 and rotor 7 are also among the rear rotors. It is due to the fact that rotor 5 and rotor 7 are in the upper layer and they are not much effected by the wake produced by front rotors, whereas rotor 6 and rotor 8 are in the lower layer and they are directly affected by the front rotors wake coming straight towards them.

Observing the front rotors of coaxial diamond and coaxial square configuration, it comes out that they do not perform as good as conventional or hybrid configurations because their rotor arrangement is such that lower layer of rotors is exactly below the upper layer causing high wake interactions at rear part of the lower layer. Rotor 2 and rotor 4 of coaxial diamond configuration are in front and exactly below rotor 1 and rotor 3, so the performance of rotor 2 and rotor 4 is much more affected as they operate inside the wake of rotor 1 and rotor 3. Lift and forward thrust of all rotors in lower layer of coaxial configurations is reduced due to this reason. Performance of rotor 6 and rotor 8 of coaxial square configuration is seriously degraded due to the fact that they not only operate inside the wake from front rotors but they also operate inside the wake of upper layer of rotors i.e. rotor 5 and rotor 7. As far as coaxial diamond configuration is concerned, their rotors all face degradation in performance because of being in coaxial arrangement but a prominent reduction in lift and forward thrust is only observed in rotor 5 and rotor 6 as these rotors are placed exactly behind the front two rotors. Fig. 26 demonstrating the ensemble-average velocity on each rotor for each octocopter configuration also confirms that conventional octocopter rotors have negligible interaction with wake whereas increment in the velocity at the back end of lower layer rotors in both coaxial configuration can also be observed. Coaxial square configuration lower rotor layer show more interaction in the form of more increment in velocity at rear part of the back rotors compared to coaxial diamond configuration. Moreover, the velocity contours of hybrid configuration demonstrates wake interaction of only rotor 6 and rotor 8 present at the back of lower rotor layer in hybrid configuration. The complete upper layer and the front two rotors in lower layer of hybrid configuration experience no wake interaction.

Table 8 FM for different octocopter configurations in steady forward flight.

Type of configuration	$FM_{combined}$ value
Conventional	0.4003
Coaxial diamond	0.3397
Coaxial square	0.3031
Hybrid	0.3635

Comparing the configurations clearly demonstrates that conventional and hybrid configurations perform much better than both coaxial configurations, as their rotors face lesser interaction with wake than coaxial configurations. Coaxial square configurations performs worst with 14.77% and 18.8% loss in lift compared to conventional and hybrid configurations. Coaxial diamond configuration shows 9.1% more lift than coaxial square configuration.

Torque magnitude also demonstrates similar trends showing more power required for rotors having more interaction with wake of neighboring rotors and less stability of vehicle. The lift and forward thrust of conventional and hybrid configurations is almost the same. Comparing coaxial square and coaxial diamond configuration for steady forward flight. It is also observed that reduction in performance of lower layer of coaxial configurations is not as much as reduction of performance of lower rotor layer in hover because in hover the wake of upper layer completely goes into the lower layer rotors, whereas in forward flight, the wake moves in oblique manner, so it interacts partially with the lower layer of rotors at the rear part configuration shows that performance of coaxial diamond configuration is much better than coaxial square.

Comparing all configurations, conventional and hybrid configurations show similar aerodynamic performance with their rotors performing approximately same as isolated rotors and better than both coaxial configurations having a significant rotor-to-rotor interaction. Table 8 shows aerodynamic

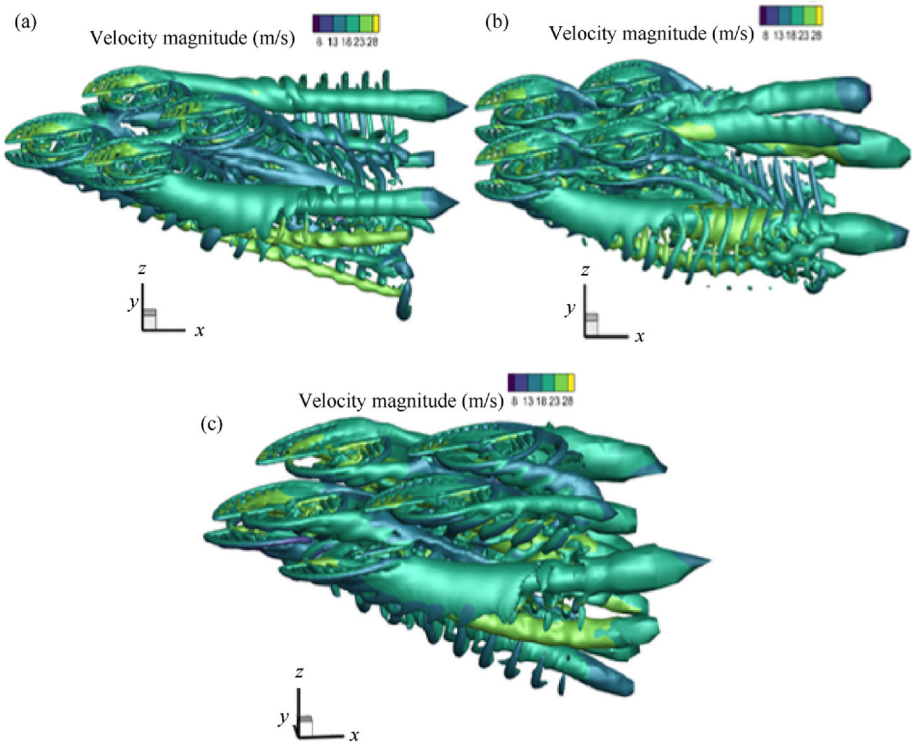


Fig. 29 Isosurface of velocity magnitude at Q criterion 5000 s^{-2} for steady forward flight of (a) coaxial diamond (b) coaxial square and (c) hybrid octocopters with bigger diameter rotors.

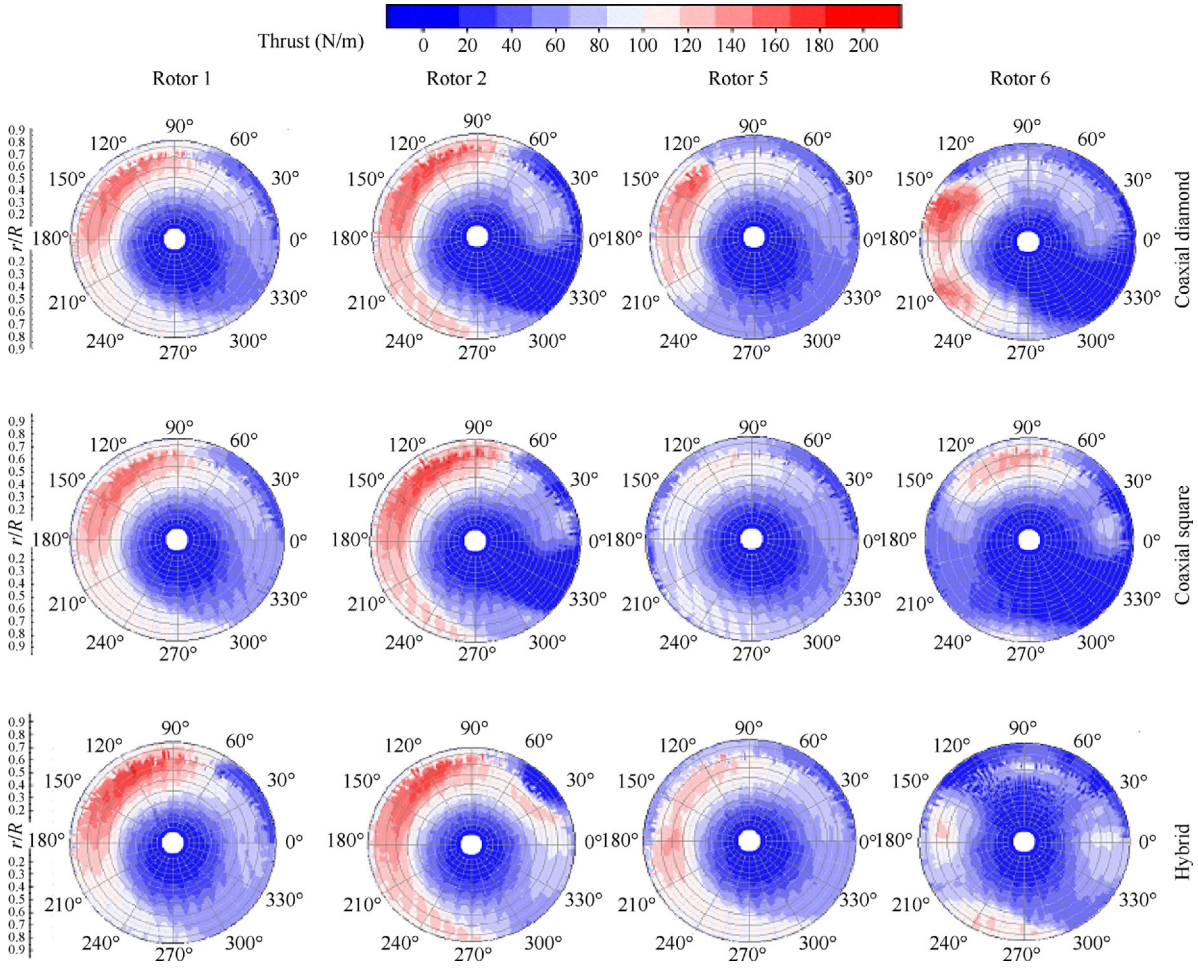


Fig. 30 Sectional thrust distribution in steady forward flight with bigger diameter rotors.

performance of each configuration in terms of their FM. It can be observed that conventional configuration has best performance and hybrid configuration is second to it, whereas coaxial square configuration performs the worst in steady forward flight.

3.3.3. Comparison of coaxial and hybrid configurations with bigger diameter rotors

For high thrust requirement missions, coaxial and hybrid configurations provide the opportunity to expand the diameter of the rotors utilizing the same confined space. Analysis for these configurations with 420 mm diameter is also presented herein. The results are for isosurface with Q criterion of 5000 s^{-1} and associated lift and thrust values are demonstrated in Fig. 29. Fig. 30 demonstrates the thrust distribution for rotors of different configurations. It can be observed that with larger diameter rotors, the impact of rotor-to-rotor interaction is increased and in addition to degradation of rotor performance due to fore rotors and upper layer of rotors, tip-to-tip interaction can also be observed clearly. Hybrid configuration depicts better thrust distribution comparatively except rotor 6 which is present exactly behind rotor 2, having to face wake interaction. Coaxial square configuration depicts worst performance of all since it has all aft rotors exactly behind fore rotors. Coaxial diamond configuration performs better than coaxial square, since its only two rotors i.e. rotors 5 and 6 are exactly behind rotors 1 and 2.

Observing the results shown in Fig. 31 for selected three configurations with bigger rotors, the trends in performance of each individual rotor is very similar to what was observed with these configurations having smaller rotors. It's just that these effects have been made more intense due to strong wake from bigger rotors. For high thrust requirement also, it is observed that hybrid configuration outperformed the other two coaxial configurations in steady forward flight as well. The performance of coaxial square configuration is worst among all confirming the same trends obtained for these configurations with smaller rotors.

Fig. 32 demonstrate the ensemble-average velocity distribution on different layers of both coaxial configurations and hybrid configuration in steady forward flight with big rotors. The upper layer rotors of all configurations experience no deterioration due to interaction with the wake, whereas lower layer rotors of all configurations are effected. It can be observed that lower layer rotors of coaxial square configuration show maximum interaction with wake, whereas all the lower layer rotors of coaxial diamond configuration are also effected but the impact is less than that of coaxial square. Observing the velocity contours of hybrid configuration confirms that the front two rotors in the lower layer has negligible interaction with wake while rotor 6 and rotor 8 at the back of lower layer are effected with maximum wake interaction seen in rotor 8 present at the top in Fig. 32(f). The overall impact of rotor wake interaction can be deduced to be minimum in hybrid configuration and maximum in coaxial square configuration.

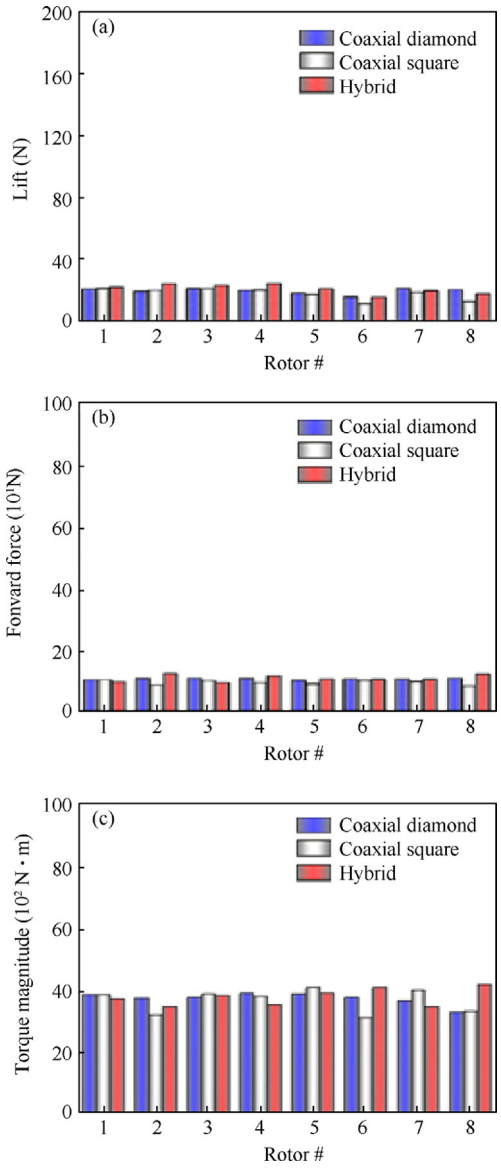


Fig. 31 Magnitude of (a) lift (b) forward force and (c) torque produced by bigger diameter rotors of coaxial and hybrid configurations in steady forward flight.

It can be observed from Table 9 that hybrid configuration outperforms the other configurations in steady forward flight with bigger diameter rotors also. Coaxial diamond is second to it, while coaxial square configuration remains as the worst configuration in aerodynamic performance.

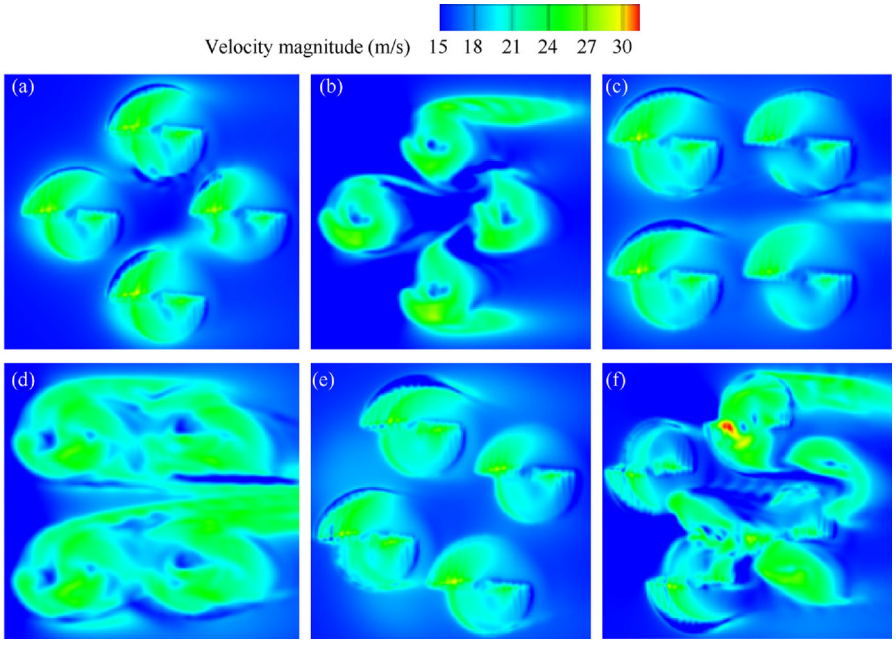


Fig. 32 Velocity magnitude on coaxial diamond rotors (a) upper layer (b) lower layer, coaxial square rotors (c) upper layer (d) lower layer and hybrid rotors (e) upper layer (f) lower layer.

Table 9 FM for different octocopter configurations in steady forward flight with bigger rotors.

Type of configuration	$FM_{combined}$ value
Coaxial diamond	0.2382
Coaxial square	0.2135
Hybrid	0.2625

4. Conclusions

(1) Results obtained using VBM method for current analysis matches with the experimental results showing the capability of VBM to predict rotor-to-rotor interaction with good accuracy and less computational cost. Therefore, VBM is a suitable selection for current study.

(2) For coaxial configuration, loading effect is dominant with $0.125R$ vertical spacing which becomes insignificant with more spacing, where wake interaction phenomenon dominates with total thrust of the configuration fairly constant. The impact of rotor-to-rotor interaction increase for bigger diameter rotors with tip-to-tip interaction also coming into play which is not apparent in normal rotors. Conventional configuration rotors behave like isolated rotors in hover.

(3) In forward flight, impact of rotor-to-rotor interaction is less than that in hover with upper rotor layer only interacting at 270°–360° azimuth of lower layer of coaxial rotor. Fore rotors wake also interact only with rear part of the aft rotors with lesser impact than coaxial rotor effect. The vortices from blades move with the flow and merge to form larger vortex. Larger rotors bring tip-to-tip interaction on half of the azimuth of each rotor with interference angle depending on rotation direction and location of rotor.

(4) In hover, coaxial configuration shows 11.89% and 14.22% loss in thrust compared to conventional and hybrid configurations with normal rotors and 15.61% loss compared to hybrid configuration with bigger diameter rotors. Coaxial square configuration performs worst in steady forward flight, with 9.1%, 14.77% and 18.8% loss in lift compared to coaxial diamond, conventional and hybrid configurations and with bigger diameter rotors, it shows loss of 17.82% and 9.96% lift compared to hybrid and coaxial diamond configurations. Loss in forward force is not very significant. The influence on torque magnitude shows slight deterioration for coaxial configurations in hover and forward flight. Combined figure of merit for each configuration in hover and forward flight confirms that hybrid configuration outperforms other configurations in overall aerodynamic performance.

CRediT authorship contribution statement

Aqib AZIZ: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yongjie SHI:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Liu YANG:** Validation, Software, Methodology. **Guohua XU:** Visualization, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. Bouabdallah S, Becker M, Siegwart R. Autonomous miniature flying robots: coming soon! - Research, development, and results. *IEEE Robot Autom Mag* 2007;**14**(3):88–98.
2. Thipphavong DP, Apaza R, Barmore B, et al. Urban air mobility airspace integration concepts and considerations. *Proceedings of the 2018 aviation technology, integration, and operations conference*. Atlanta, Georgia. Reston: AIAA; 2018.
3. Zhang H, Zou Y, Zhang Q, et al. Future urban air mobility management: Review. *Acta Aeronaut Astronaut Sin* 2021;**42**(7) 024638 [Chinese].
4. Warwick G. New Zealand welcomes flight tests of Kitty Hawk's eVTOL air taxi: Full-scale prototypes of Cora air taxi in flight testing; transitional eVTOL combines rotors for vertical flight with wings for efficient forward flight. Aviation Week & Space Technology; [updated 2018 Mar 26; cited 2023 Dec 9]. Available from: https://aviationweek.com/aviation-week-space-technology/new-zealand-welcomes-flight-tests-kitty-hawks-evtol-air-taxi.
5. Rajendran S, Srinivas S. Air taxi service for urban mobility: A critical review of recent developments, future challenges, and opportunities. *Transp Res Part E Logist Transp Rev* 2020;**143**:102090.
6. Higgins RJ, Barakos GN, Shahpar S, et al. A computational fluid dynamic acoustic investigation of a tiltwing eVTOL concept aircraft. *Aerosp Sci Technol* 2021;**111**:106571.
7. Holden J, Goel N. Fast-forwarding to a future of on-demand urban air transportation [Internet]. San Francisco, CA: UBER; [updated 2016 Oct 27; cited 2023 Dec 9]. Available from: https://evtol.news/__media/PDFs/UberElevateWhitePaperOct2016.pdf.
8. Joshi PM. Wing analysis of a flapping wing unmanned aerial vehicle using CFD. *Int J Adv Eng Res Dev* 2015;**2**(5):212580058.
9. Houghton J, Hoburg W. Fly-by-wire control of a monocopter. Cambridge: Massachusetts Institute of Technology; 2008.
10. Otsuka H, Nagatani K. Thrust loss saving design of overlapping rotor arrangement on small multirotor unmanned aerial vehicles. *2016 IEEE international conference on robotics and automation (ICRA)*; Stockholm, Sweden. Piscataway: IEEE; 2016. p. 3242–8.
11. Nugroho G, Hutagaol YD, Zuliardiansyah G. Aerodynamic performance analysis of VTOL arm configurations of a VTOL plane UAV using a computational fluid dynamics simulation. *Drones* 2022;**6**(12):392.
12. Chahba S, Sehab R, Morel C, et al. Fast sizing methodology and assessment of energy storage configuration on the flight time of a multirotor aerial vehicle. *Aerospace* 2023;**10**(5):425.
13. Stokkermans T, Veldhuis L, Soemarwoto B, et al. Breakdown of aerodynamic interactions for the lateral rotors on a compound helicopter. *Aerosp Sci Technol* 2020;**101**:105845.
14. Onen AS, Senipek M, Mutlu T, et al. Modelling and control of a quad-rotor robot. *ICUAS*. 2015.
15. Magnussen Ø, Hovland G, Ottestad M, Multicopter UAV design optimization. *2014 IEEE/ASME 10th international conference on mechatronic and embedded systems and applications (MESA)*; Senigallia, Italy. Piscataway:IEEE; 2014. p.1–6.
16. Er MJ, Yuan S, Wang N, et al. Modeling and controller design of a VTOL UAV. *ICCA*. 2013.
17. Sinha P, Esden-Tempski P, Forrette CA, et al. Versatile, modular, extensible VTOL aerial platform with autonomous flight mode transitions. *2012 IEEE aerospace conference*; Big Sky, MT, USA. Piscataway: IEEE; 2012. p. 1–17.
18. Ranganathan T, Thondiyath A, Kumar SPS. Design and analysis of an underwater quadrotor - AQUAD. *2015 IEEE underwater technology (UT)*; Chennai, India. Piscataway: IEEE; 2015.p.1–5.
19. Pounds P, Mahony R. Design principles of large quadrotors for practical applications. *2009 IEEE international conference on robotics and automation*. Kobe, Japan. Piscataway: IEEE; 2009. p. 3265–70.
20. Ko J, Lee S. Quantification of wake interaction effects on multi-rotor configurations in forward flight. *Aerosp Sci Technol* 2023;**135**:108188.
21. Lakshminarayan VK, Baeder JD. High-resolution computational investigation of trimmed coaxial rotor aerodynamics in hover. *J Am Helicopter Soc* 2009;**54**(4):042008.
22. Yoon S, Lee HC, Pulliam TH. Computational analysis of multi-rotor flows. *Proceedings of the 54th AIAA aerospace sciences meeting*; San Diego, California, USA. Reston: AIAA; 2016.
23. Passe BJ, Sridharan A, Baeder JD. Computational investigation of coaxial rotor interactional aerodynamics in steady forward flight. *Proceedings of the 33rd AIAA applied aerodynamics conference*; Dallas, TX. Reston: AIAA; 2015.
24. Hayami K, Sugawara H, Yumino T, et al. CFD analysis on the performance of a coaxial rotor with lift offset at high advance ratios. *Aerosp Sci Technol* 2023;**135**:108194.
25. Benedict M, Winslow J, Hasnain Z, et al. Experimental investigation of micro air vehicle scale helicopter rotor in hover. *Int J Micro Air Veh* 2015;**7**(3):231–55.
26. Zhang WW, Xu B, Zhang HT, et al. Analysis of aerodynamic characteristics of propeller systems based on Martian atmospheric environment. *Drones* 2023;**7**(6):397.
27. Russell CR, Jung J, Willink G, et al. Wind tunnel and hover performance test results for multicopter UAS vehicles. *American Helicopter Society (AHS) international annual forum and technology display*. 2016.
28. Tugnoli M, Montagnani D, Syal M, et al. Mid-fidelity approach to aerodynamic simulations of unconventional VTOL aircraft configurations. *Aerosp Sci Technol* 2021;**115**:106804.
29. Caprace DG, Ning A, Chatelain P, et al. Effects of rotor-airframe interaction on the aeromechanics and wake of a quadcopter in forward flight. *Aerosp Sci Technol* 2022;**130**:107899.
30. Zhang T, Barakos GN, Foster M, et al. High-fidelity aerodynamic and acoustic design and analysis of a heavy-lift eVTOL. *Aerosp Sci Technol* 2023;**137**:108307.
31. Yang YN , Liu Y , Li Y , et al. Aerodynamic and aeroacoustic characteristics of a multicopter propeller during forward flight. *Proceedings of the 2018 joint propulsion conference*; Cincinnati, Ohio. Reston: AIAA; 2018.
32. Shukla D, Komerath N. Multirotor drone aerodynamic interaction investigation. *Drones* 2018;**2**(4):43.
33. Prewitt NC, Belk DM, Shyy W. Parallel computing of overset grids for aerodynamic problems with moving objects. *Prog Aerosp Sci* 2000;**36**(2):117–72.

34. Chan WM, Van Dalsem WR. Hyperbolic methods for surface and field grid generation. Moffett Field(CA): Ames Research Center; 1996.
35. Meakin RL. Composite overset structured grids. In: Thompson JF, Soni BK, Weatherill NP, editors. *Handbook of grid generation*. Calabas Municipality: CRC Press; 1999. p. 1–20 [chapter 11].
36. Zori LAJ, Rajagopalan RG. Navier—Stokes calculations of rotor—airframe interaction in forward flight. *J Am Helicopter Soc* 1995;**40**(2):57–67.
37. Burton T, Sharpe D, Jenkins N, et al. *Wind energy handbook*. Oxford (UK): John Wiley & Sons; 2001.
38. Zhou WW, Ning Z, Li HX, et al. An experimental investigation on rotor-to-rotor interactions of small UAV propellers. *Proceedings of the 35th AIAA applied aerodynamics conference*; Denver, Colorado. Reston: AIAA; 2017.
39. Ning Z. Experimental investigations on the aerodynamic and aeroacoustics characteristics of small UAS propellers [dissertation]. Ames: Iowa State University; 2018.
40. Lakshminarayan VK, Baeder JD. Computational investigation of microscale coaxial-rotor aerodynamics in hover. *J Aircr* 2010;**47**(3):940–55.
41. Park SH, Kwon OJ. Numerical study about aerodynamic interaction for coaxial rotor blades. *Int J Aeronaut Space Sci* 2021;**22**(2):277–86.
42. Barbely N, Komerath N. Coaxial rotor flow phenomena in forward flight. *SAE aerospace systems and technology conference*; Hartford, CT, USA.2016.
43. Ventura Diaz P, Yoon S. High-fidelity computational aerodynamics of multi-rotor unmanned aerial vehicles. *Proceedings of the 2018 AIAA aerospace sciences meeting*; Kissimmee, Florida. Reston: AIAA; 2018.